



**Verified Carbon
Standard**

YILMAZLAR BIOGAS POWER PLANT PROJECT



Document Prepared by Life İklim ve Enerji Ltd. Şti.

Project Title	Yılmazlar Biogas Power Plant Project
Version	04
Date of Issue	12-January-2024
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Yılmazlar Madencilik Sanayi ve Ticaret Limited Şirketi (hereinafter mentioned as Project Proponent) has constructed a cattle farm with the name of Kayı Farm located in Derebent Village of Tavşanlı District of Kütahya Province in Türkiye. The Project Proponent has been operating the farm since 2013. As a result of farm activities, some amount of manure is being generated. To treat this waste, Project Proponent invested in Yılmazlar Biogas Power Plant Project (hereinafter mentioned as the Project), and the construction of the plant was completed on June 2021 and the relevant provisional acceptance protocol has been taken. However, the operation of the project activity, i.e. the start date of GHG emission reduction, was started on 29-September-2021 as per the SCADA values of the flow meters (which are located after the anaerobic digesters and before the flare). Thanks to the Project, the manure resulting from the farm activities can be disposed of safely and environmentally friendly, and clean energy can be generated.

The Project has been designed to have a lifetime of 49 years in accordance with generation license. It was planned that cattle manure from Kayı Farm and chicken litter from nearby farms would be mixed and digested to produce biogas by anaerobic digestion and then converted to electricity biogas power plant. The remaining digestate from the anaerobic digesters is treated further to produce organic fertiliser.

On 20 May 2020, EMRA granted the Project activity with a Generation License of 1.560 MWe and 11,700 MWh/year for 49 years. The license is granted for the Plant to have 1 gas engine with a capacity of 1.560 MWe. The provisional acceptance date of the plant is 27 June 2021. The construction of the power plant was started in February 2020 after signing the Construction Agreement with the contractor company and then on 09 June 2021, the connection agreement was signed with EPIAS. On the other hand, the electricity generation was started on 29-September-2021 as per the SCADA values of the flow meters.

Prior to the start of the project activity (in the baseline scenario) manure from the farms were being washed out from underneath the animal barns and stored in anaerobic lagoons which allowed the manure to decay and emit methane to the atmosphere along with several other toxic gases. Such applications also bear the risk of lagoon overflow due to increased rainfalls, strong Biogas or improper construction.

The environmental, socio-economic, and technological benefits of the Project are described as follows:

- Reduction in fossil fuel use (imported or local) by using renewable energy resources,
- Reduction in greenhouse gas emissions from the animal livestock by using biogas for electricity production,

- Reduction in energy production costs and imported energy amounts,
- Improvement of environmental conditions (GHG and odour) and safety in the area.

As per the feasibility study, the amount of receiving daily animal manure is around 125 tonnes.

The project is expected to reduce 35,855 tCO₂e/year in average in the first 7-year crediting period between commission date of 29-September-2021 and ending date of 28-September-2028.

Audit history of the project is given in the table below.

<u>Audit Type</u>	<u>Period</u>	<u>Program</u>	<u>VVB Name</u>	<u>Number of years</u>
Validation	29-September-2021 – 28 September 2028	VCS	Earthood Services Private Limited	7 years
First Verification	29-September-2021 – 31-March-2023	VCS	Earthood Services Private Limited	1 years and 6 months
Total	29-September-2021 – 31-March-2023	VCS	Earthood Services Private Limited	1 years and 6 months

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 Renewable Energy

Sectoral scope: 13 Waste Handling and Disposal

Type I: Renewable Energy Projects

Type III: Other project activities

The Project is neither an AFOLU nor a grouped project

1.3 Project Eligibility

The scope of the VCS Program includes:

- 1) The seven Kyoto Protocol greenhouse gases: The project is expected to avoid Methane (CH₄) emissions from the anaerobic animal manure management system in the baseline scenario, which will be captured and destroyed in the project scenario as well as renewable energy generation. Thus, the project applicable to this scope.
- 2) Ozone-depleting substances: Not Applicable.
- 3) Project activities supported by a methodology approved under the VCS Program through the methodology development and review process: Not Applicable.

- 4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded: The applied methodologies AMS-III.D (Version 21.0) and AMS-I.D. (Version 18.0) are methodologies approved under CDM Program, which is a VCS approved GHG program.
- 5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements: Not Applicable.

The biogas generated from the anaerobic digesters is recovered and utilized as an energy source within the project boundary. Apparently, the project does not generate GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.

Furthermore, the project does not belong to the project activities excluded in Table 1 of VCS Standard 4.4.

Thus, the project is eligible under the scope of VCS program.

1.4 Project Design

- ☒ The Project includes a single location or installation only
- ☐ The Project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☐ The Project is a grouped project

1.5 Eligibility Criteria

The Project is not a grouped project. The Project belongs to one project proponent. In the scope of this Project, only one location will be considered for emission reductions.

1.6 Project Proponent

Organisation name	Yılmazlar Madencilik Sanayi ve Ticaret Limited Şirketi
Contact person	Mr Ramazan Fidan
Title	Biogas Plant Operation Head
Address	Tunçbilek Karayolu 7. Km Tavşanlı/Kütahya/Türkiye
Telephone	+90 (274) 600 02 00
Email	r.fidan@yilmazlarmadencilik.com.tr

1.6 Other Entities Involved in the Project

Organisation name	Life İklim ve Enerji Ltd. Şti.
Role in the Project	Consultant for Carbon Certification
Contact person	Mr. Ramazan Aslan
Title	General Manager
Address	Oğuzlar Mah. 1377. Sk. No:19/9 06520, Balgat, Çankaya/Ankara
Telephone	+90 (312) 481 21 42
Email	ramazan.aslan@lifeenerji.com

1.7 Ownership

Ownership of the Yılmazlar Biogas Power Plant project belongs to Yılmazlar Madencilik San. ve Tic. Ltd. Şti., and all emission reductions/removals are granted to the company. The tax plate as proof of operating certification of the company given by the relevant tax office of the host country is provided to the validating VVB as proof of the Title.

1.8 Project Start Date

The construction of the plant was completed on June 2021 and the relevant provisional acceptance protocol has been taken. However, the operation of the project activity, i.e. the start date of GHG emission reduction, was started on 29-September-2021 as per the SCADA values of the flow meters (which are located after the anaerobic digesters and before the flare). Hence, the start date of the project activity is 29-September-2021.

1.9 Project Crediting Period

The Project crediting period for this Project is 7 years, renewable twice.

First Crediting Period Start Date: 29 September 2021

First Crediting Period End Date: 28 September 2028

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the Project are:

☐ <20,000 tCO₂e/year

☒ 20,000 – 100,000 tCO₂e/year

☐ 100,001 – 1,000,000 tCO₂e/year

☐ >1,000,000 tCO₂e/year

Project Scale	
Project	X
Large Project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
29 th September 2021 – 31 st December 2021	9,135
01 st January 2022 - 31 st December 2022	35,855
01 st January 2023 - 31 st December 2023	35,855
01 st January 2024 - 31 st December 2024	35,855
01 st January 2025 - 31 st December 2025	35,855
01 st January 2026 - 31 st December 2026	35,855
01 st January 2027 - 31 st December 2027	35,855
01 st January 2028 - 28 th September 2028	26,718
Total estimated ERs	250,983
Total number of crediting years	7
Average annual ERs	35,855

1.11 Description of the Project Activity

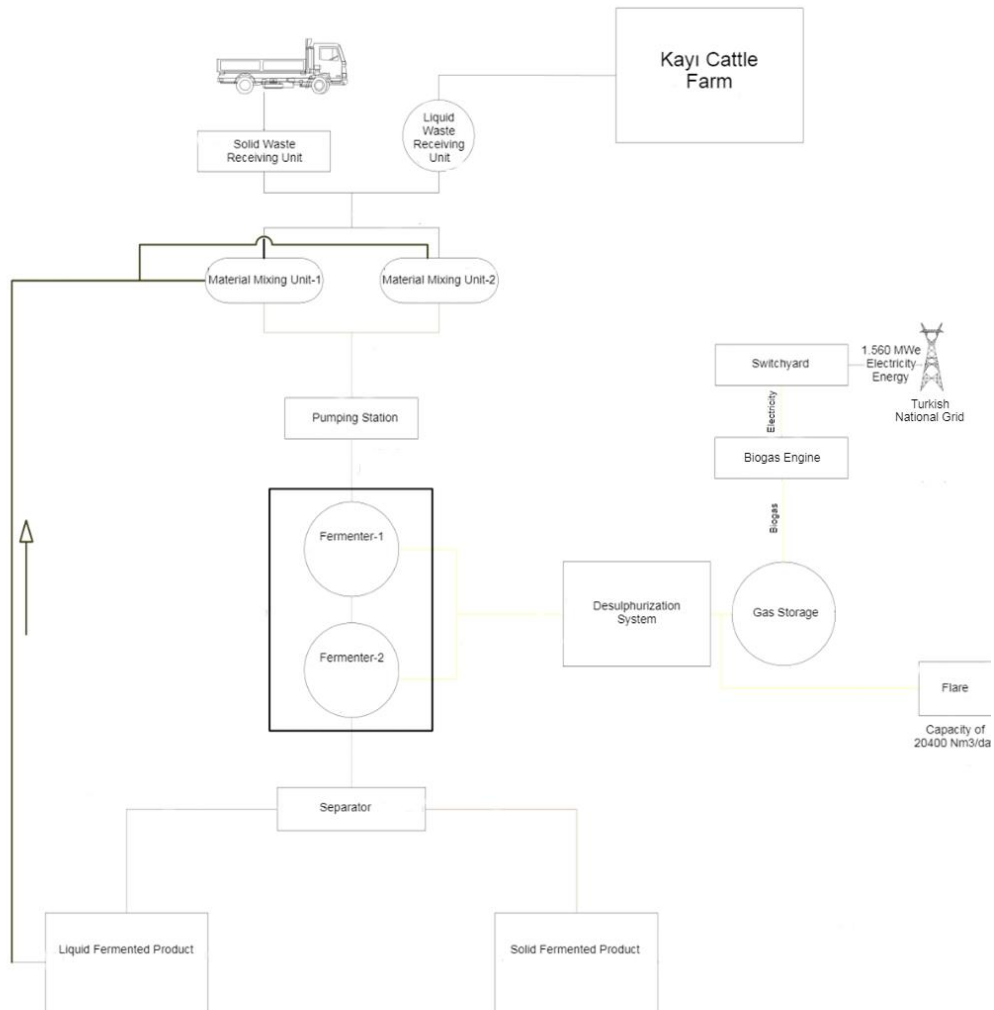
As mentioned before, two sources of waste are being digested in the Yilmazlar Biogas Power Plant. The cattle manure comes from the Kayı Farm via pipes where Yilmazlar Biogas Power Plant is. Ownership of Yilmazlar BPP and Kayı Farm belongs to same Project Proponent. On the other hand, chicken litter is brought from poultry farms by trucks.

Yilmazlar Biogas Power Plan is designed to include units listed below:

- Solid Waste Receiving Unit
- Liquid Waste Receiving Unit
- Material Mixing Unit
- Anaerobic Digesters

- Desulphurization and Gas Storage
- Separator Units
- Gas Engine and Flare Units

Process diagram of the plant is given in the figure below.



Solid Waste Receiving Units

Chicken litter supplied from other farms is first taken to the solid waste receiving unit. The solid waste receiving unit is made of at least 30 cm thick, C30/C37 impermeable concrete and non-combustible material, so it has a leakproof base and closed top and sides.

The purpose of the solid waste receiving unit is to empty the chicken litter of the vehicles arriving at the facility as soon as possible and to store the wastes appropriately throughout the year. Solid wastes in the unit are taken to the material mixing unit in the amount to be used daily.

The area of the solid waste receiving unit is 300 m². The base of the unit was designed to collect leachate from the bottom to prevent contamination of any water body, and collected leachate is directed to the material mixing unit.

Liquid Waste Receiving Units

All liquid waste generated and received by the plant is animal manure produced in Kayı Farm where Yilmazlar Biogas Power Plant is actually in. Liquid wastes are collected from the farm and pumped to the power plant by pipeline.

Liquid wastes are firstly taken to the liquid waste receiving unit. The liquid waste receiving unit is made of at least 30 cm thick, C30/C37 impermeable concrete and non-combustible material, so that it has leakproof base and closed top and sides.

Mixing Unit

Solid and liquid waste is mixed daily in two mixing units. The aim of having two material units is when one unit is in mixing process, the other one pumps the waste into digesters. Therefore, when one unit gets empty, the other will be ready to pump. Consequently, the non-stop feeding of digesters is maintained.

Each material mixing unit has an 84 m³ capacity (168 m³ total). To have a homogenous mixture, each of them has a submersible mixer. Moreover, they are covered from top to prevent odour.

Solid and liquid wastes accepted by the plant and is mixed in order to provide suitable moisture for digestion. Material mixing units are made of at least 30 cm thick, C30/C37 impermeable concrete and non-combustible material so that it has a leakproof base and closed top and sides.

Anaerobic Digesters

There are two anaerobic digesters in the biogas plant with 6430 m³ capacity each. The reactors are made from reinforced concrete, leak-proof, and designed to have a self-heating and mixing system. The waste mixture that has suitable moisture is digested by microorganisms under anaerobic conditions and converted into carbon dioxide and methane gas. In the digester, methane gas is produced under mesophilic conditions. The methane gas produced is withdrawn by blowers from the digester and delivered to the biogas engine unit to obtain electrical energy. Hot water pipes placed inside the digesters provide automatic control of the temperature of the digester to keep it constant with continuous hot water circulation.

Also, mixers inside the digesters are activated according to the time parameters determined in the Plant's automation system to keep the mixture homogenous.

Before starting the Project (in the baseline scenario), manure from the farms was being washed out from underneath the animal barns and stored in anaerobic lagoons, which allowed the

manure to decay and emit methane into the atmosphere along with several other toxic gases. Such applications also bear the risk of lagoon overflow due to increased rainfalls, strong Biogas or improper construction.

Energy requirements for the Plant will be supplied from the national grid, and a generator is ready to use in an emergency electricity need.

The Project is estimated to generate a net 11,700 MWh/y of electricity in accordance with generation license, while the biogas production capacity will be around 6,275,335 m³/y

Desulphurisation Unit and Gas Storage

Hydrogen sulphide, a highly corrosive acidic gas contained in the biogas, and water vapour are also eliminated in the desulphurisation unit employed in the biogas plant. Some amount of hydrogen sulphide (H₂S) in the biogas is reduced by addition of oxygen to the digesters. After that product gas is taken to desulphurisation unit and hydrogen sulphide is reduced to desired level. The produced biogas is stored in double membrane inflatable roofs and then accelerated by a blower system to the gas engine.

The biogas stored in the gas balloon is sucked from this area by means of gas blowers. The biogas, which is compressed and pressurized by the gas blowers, is transported to the inlet line of the biogas engine system at the desired pressure.

Gas Engine and Flare Unit

The gas engine unit has one electricity-generating engine with a capacity of 1.56 MWe. Biogas energy fed to gas engines will be converted into electricity. All the electricity produced is supplied to the grid, which also meets the electricity consumed by the Plant. There is also a flare unit with a capacity of 850 m³/hr in the biogas plant for emergencies where the engines don't combust produced biogas. The combustion temperature of the flare is around 1200°C.

Dewatering (Separator)

Two separators are employed to separate solid and liquid phases of digestate. Separators can separate digestate up to 25% dry matter and 95% liquid matter. The area of the separator unit is 50 m².

The monitoring equipment (i.e. gas analyser, weighbridge and electricity meters) is in place at the project site to monitor relevant ex-post parameters. The technical details of the monitoring equipment are as follows:

Monitoring Equipment	Brand	Serial Number	Calibration Date	Location
Weighbridge	YİĞİTOĞLU/SER 12	221	22.03.2018	Entrance of the project site

Flow meter	Prosonic Flow 200	S2195302000	19.03.2021	After the anaerobic digesters
Gas Analyzer-1	SWG100 BIOGAS	081392	15.06.2023	In the blower room
Electricity Meter - Main	80281239	Makel - C520 AMT.2556	25.06.2021	After the gas engine
Electricity Meter - Back-up	80281229	Makel - C520 AMT.2556	25.06.2021	After the gas engine

Since baseline emissions from the manure treatment processes are calculated using electricity generation value, not flow of the biogas, the flow meter before the gas engine is not taken into consideration as a monitoring equipment. For the flare unit, manual measuring system is in place since the flare works only at emergency cases. Nevertheless, a cross-checked method is used for the relevant flow of the biogas to ensure that project emissions are measured correctly. The cross-checked method is as follows:

$$\text{Flow of the biogas to flare} = \text{Capacity of the flare (Nm}^3\text{/h)} \times \text{Operating time of the flare (h)}$$

During the current monitoring period, the flare was operated between May 2022-March 2023 when emergency cases occurred.

1.12 Project Location

The Project is located in Derbent Village of Tavşanlı District of Kütahya Province in Türkiye. Total area of the project is 24.330.18 m². Derbent village is 1250 meters north of the project, Yeniköy village is 2120 meters south, and Demirbilek Village is 3500 meters west which are the nearest settlements.

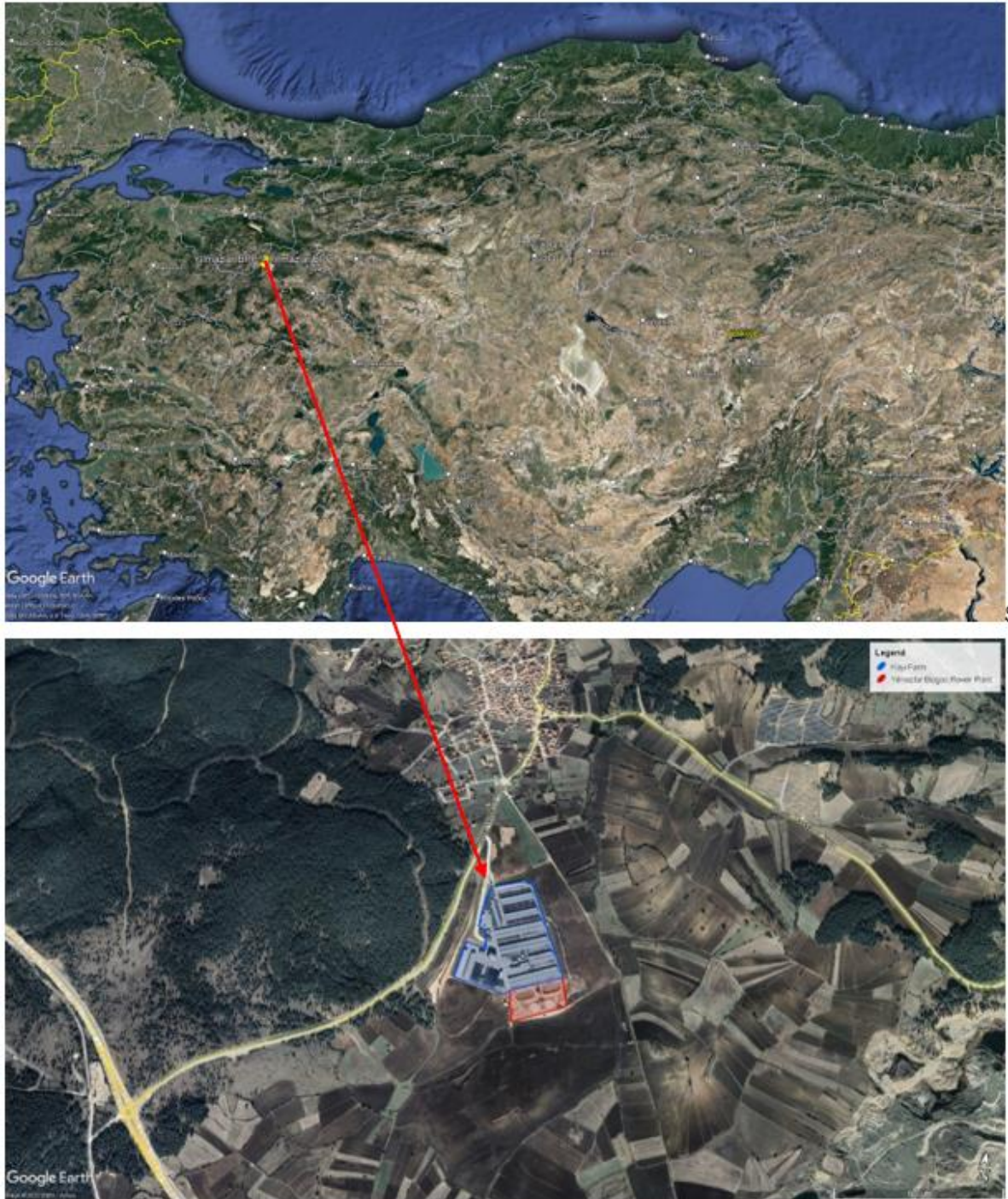


Figure 1: Satellite View of Yilmazlar Biogas Power Plant

The coordinates of the plant is given in the table below.¹

¹ Generation License

E	N
29.353777222303226	39.628529407842706

The farms of the manure supplying are also in the project boundary. The farms and their coordinates are as follows:

Farms	Coordinates
Kayı Farm	29° 21'9.39" E ; 39° 37'42.19" N
As Tarım	29° 28'14.44" E ; 39° 32'50.51" N
Baş Tarım	29° 27'16.50" E ; 39° 30'37.70" N
Özkantar Gıda	29° 29'2.92" E ; 39° 30'56.56" N
Halil Yeşil Yumurta Üretim	29° 28'38.77" E ; 39° 31'18.85" N
Yoncalı Gıda	29° 26'14.36" E ; 39° 31'23.64" N
Yumurtalar Tarım	29° 28'42.65" E ; 39° 31'19.31" N
Emre Kayner	29° 28'35.76" E ; 39° 31'16.43" N
Halil Küçük	29° 28'39.85" E ; 39° 31'18.85" N
Saydam Tarım	29° 28'35.04" E ; 39° 31'21.16" N
Bayraktar	29° 26'17.29" E ; 39° 31'19.17" N
Ali Aydın	28° 47'20.71" E ; 39° 51'25.86" N

1.13 Conditions Prior to Project Initiation

The alternative scenarios of animal manure in the absence and prior to the Project include:

- Electricity would be generated in existing grid-connected electricity plants
- The animal manure would be treated in the anaerobic lagoons

Before the Project, cattle and chicken litter generated from cattle breeding and poultry farms were left to decay in anaerobic lagoons. Therefore, greenhouse gases generated from the manure would be directly released into the atmosphere. Moreover, the electricity generated by the Project would be replaced by the power plants connected to the national grid, and fossil fuel-fired boilers would supply the thermal energy.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

Turkish legislation on manure management concentrates on control of the manure waste, and its potential threat on air, water and soil pollution.

The related legislation is presented below:

Relevant Law and Regulations for the Project	Comment
“Environmental Law ² ” dated 26/04/2006 numbered 2872	The law prohibits the discharge, storage, transfer, and disposal of any kind of waste that does not meet the requirements according to Turkish regulations and standards. It includes the “polluter pays principle” and ensures that the producers take necessary measures to minimise waste. The law doesn’t mandate any alternative treatment processes for the animal manure management.
Communiqué on Fermented Product Management by Mechanical Separation, Biodrying, Biomethanization dated 10/10/2015 numbered 29498	The purpose of this Communiqué is to ensure the management of biodegradable wastes without harming the environment and human health, to reduce the amount to be disposed of in landfills, to the technical criteria of mechanical separation, biodrying and biomethanization facilities with material or energy recovery facilities, to the management of the fermented product obtained in biomethanization facilities.
Regulation on Control of Odor Generating Emissions ³	The purpose of this Regulation is to regulate the administrative and technical procedures and principles for the control and reduction of odor-causing emissions.
Electricity Market Law ⁴ dated 20/02/2001 and numbered 4628	The Law aims to ensure the development of a financially sound and transparent electricity market operating in a competitive environment under civil law provisions and the delivery of sufficient, good quality, low cost and environment-friendly electricity to consumers and to ensure the autonomous regulation and supervision of this market. The regulation doesn’t mandate any alternative treatment processes for electricity production.
Law on Utilisation of Renewable Energy Resources for the Purpose of Generating Electricity Energy ⁵ dated 10/05/2005 and numbered 5346	The purpose of this Law is to expand the utilisation of renewable energy resources for generating electrical energy, to benefit from these resources in a secure, economical and qualified manner, to increase the diversification of energy resources, to reduce greenhouse gas emissions, to assess waste products, to protect the

² See: <https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=2872&MevzuatTur=1&MevzuatTertip=5>

³ See: <https://www.resmigazete.gov.tr/eskiler/2010/09/20100904-10.htm>

⁴ See: <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.4628-20080709.pdf>

⁵ See: <https://www.mevzuat.gov.tr/MevzuatMetin/1.5.5346.pdf>

	environment and to develop the related manufacturing sector for realising these objectives. The Law brings an incentive of 13.3 \$ cent/kWh for the electricity production from biomass. It also brings incentives for local equipment purchases such as turbines, engines, cogeneration systems etc. The regulation does not mandate any alternative treatment processes for renewable electricity production.
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1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The Project has neither been registered nor seeking registration under any other GHG programs. The Project is seeking registration only in the VCS program.

1.15.2 Projects Rejected by Other GHG Programs

The Project didn't apply for registration in any other GHG program and was not rejected by any other GHG programme.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the Project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

There are no emissions trading programs or any other mechanism in Türkiye that includes GHG allowance trading in Türkiye; subsequently, the Project does not fall under any such mechanisms.

1.16.2 Other Forms of Environmental Credit

Has the Project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

The Project has not sought or received any other form of GHG-related environmental credits.

1.16.3 Supply Chain (Scope 3) Emissions

The project will remove methane emissions by destroying methane and will cause emission reduction by electricity supplying to the national grid. Apart from that Project will not impact on any scope 3 emissions.

1.17 Sustainable Development Contributions

1.17.1 Sustainable Development Contributions Activity Description

SDG 13 Climate Action

The project will contribute SDG 13 Climate action and indicator 13.2.1 “Number of countries with nationally determined contributions, long-term strategies, national adaptation plans and adaptation communications, as reported to the secretariat of the United Nations Framework Convention on Climate Change”. The Project Activity results in emissions reduction by capturing the methane in cattle and chicken litter and converting it to electricity in the biogas power plant. Without the Project, the manure generated in the farms would be left to decay in an anaerobic lagoon producing methane gas. With the Project, the collected animal manure is being used in the digesters to produce biogas and energy, avoiding the release of methane into the atmosphere along with several other toxic gases. By avoiding methane emissions and generating renewable energy, the Project will result in an average emission reduction of 35,855 tCO₂e annually. Additionally, in the biogas plant, captured methane is converted to electricity, where the generated electricity is supplied to the grid. Without the Project, the electricity generated would be supplied from the national grid where thermal power plants are connected, causing more GHG emissions.

SDG 7 Affordable and Clean Energy

The Project results in renewable energy generation by digesting animal manure in an anaerobic environment to produce electricity. The baseline scenario for using gas generated from an anaerobic digester (biogas) is that the electricity delivered by the Project to the National Grid would have otherwise been generated by the operation of grid-connected power plants and/or by the addition of new generation sources into the grid. Turkish electricity generation is mainly composed of thermal power plants, and the share of fossil fuel resources in increasing energy demand of the country; has also increased significantly in recent years. In the absence of the proposed Project, the same amount of electricity would be required to be supplied via either the existing power plants or by increasing the number of thermal power plants, thus this would cause increase in GHG emissions.

The clean energy generated by the Project is calculated based on the electricity generated per annum. The Project is expected to generate 11,700 MWh of clean energy annually and contributes to the share of renewable energy sources.

In order to estimate the outcome of SDG 7, Indicator 7.2.1, and “Renewable energy share in the total final energy consumption” of SDG Target 7.2: “By 2030, increase the share of renewable energy in the global energy mix substantially” can be employed. The Project will increase renewable energy share in the country’s energy mix.

The Project’s contribution to SDG 7 is determined via a simple calculation that provides the overall renewable energy generation fed into the grid. This approach is based on the following calculation:

The quantity of net electricity generation supplied by the project plant to the grid in the related year has been calculated by subtracting “Electricity consumption from the grid (MWh)” from “Electricity supplied to the grid (MWh)”

Calculation of Electricity Generation of Yilmazlar BPP:

Net Electricity Generation of Yilmazlar BPP = Electricity supplied to the grid (MWh) - Electricity consumption from the grid (MWh)

SDG 8 Decent Work and Economic Growth

Türkiye has a high rate of unemployment. In the Project, 16 people have been employed and more than half of them are local, and all employees is trained at least once a year. Without this Project, there would be no employment and no training. Therefore, it would not be possible to contribute to the project region's economy and the people's education without the Project. Health and safety (HSE) training certificates for all employees are provided, and all employees are to be trained according to the requirements. The Project promotes inclusive and sustainable economic growth, employment and decent work for all and contributes to the decrease in the unemployment rate. Therefore, the project will contribute SDG 8, and indicators 8.5.1 “Average hourly earnings of female and male employees, by occupation, age and persons with disabilities” and 8.8.2 “Level of national compliance with labour rights (freedom of association and collective bargaining) based on International Labour Organization (ILO) textual sources and national legislation, by sex and migrant status”.

1.17.2 Sustainable Development Contributions Activity Monitoring

SDG13: Climate Action

The emission reduction parameter can analyse this contribution. The Project's annual average emissions reduction is estimated to be 35,855 tCO₂e for this crediting period.

SDG8: Decent Work and Economic Growth

The Project has employed 16 workers. The employment of local people with the necessary technical qualifications for the required post has been the priority and enhanced by all project activities during the construction and operation of the Yilmazlar BPP. As a result, increased job opportunities and project business activities have partially eliminated local poverty and unemployment. Construction materials for the foundations, cables, and other auxiliary equipment have been preferentially sourced locally. Moreover, as a contribution of the Project to

the region's welfare, the quality of the Electricity consumed in the area has been increased by local electricity production, contributing to decreasing distribution losses.

Moreover, Project improves the quality of employment by training the employee. Thus, a series of training will be organised with different focuses such as health and safety, environmental management and technical capacity building within the power plant.

SDG7: Renewable Affordable and Clean Energy

The Project's gross annual renewable electricity generation has been estimated to be 81,900 MWh during this Crediting Period.

In the baseline scenario, cattle manure generated from cattle breeding was left to decay in anaerobic lagoons. Therefore, greenhouse gases generated from the manure would be released into the atmosphere. The baseline scenario for the use of gas generated from a biogas power plant is that the electricity delivered by the Project to the National Grid would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid. Turkish electricity generation is mainly composed of thermal power plants, and the share of renewable resources⁶; has decreased significantly in recent years. There is an increasing demand for electricity, which is expected to continue in the foreseeable future. In the absence of the proposed Project, the same amount of electricity is required to be supplied via either the existing power plants or by increasing the number of thermal power plants, thus increasing GHG emissions.

⁶ Please see <https://webapi.teias.gov.tr/file/1bfb54a8-26c8-4e65-b354-f47044a1bf6b?download>

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By destruction of methane and utilizing it to generate renewable energy, 37,150 tCO ₂ e emission reduction has been provided during the monitoring period.	By destruction of methane and utilizing it to generate renewable energy, 37,150 tCO ₂ e emission reduction has been provided during life time of the project .
2)	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities	Implemented activities to increase	During the monitoring period, total of 16 workers has been employed in this monitoring period.	Since the operation start date of the Plant, the Project has employed a total of 16 workers.
3)	8.8	8.8.2 Level of national compliance with labour rights (freedom of association and collective bargaining) based on International Labour Organization (ILO) textual sources and national legislation, by sex and migrant status	Implemented activities to increase	During the monitoring period, the Project provides a total of 1 training to all employees.	Since the operation start date of the Project, the Project has provided a total of 1 training to all employees.

4)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	By utilizing methane in biogas engine, total of 8,992 MWh renewable energy has been generated during the monitoring period.	By utilizing methane in biogas engine, total of 8,992 MWh renewable energy has been generated during the life time of the project.
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1.18 Additional Information Relevant to the Project

Leakage Management

The floor of the waste reception unit is made of 30 cm thick, C30/37 concrete and non-flammable material to ensure its impermeability. A separate collection mechanism is constructed at the base of the waste reception unit to prevent the waste from contacting or leaking with groundwater, sewage or surface water. The ground is sloped appropriately to collect the leachate in the waste reception area. Also, an impervious concrete floor is applied for leachate on the Material Recovery Facility floor, and grate channels are used to collect the leachate. Collected leachate is given to material mixing unit.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

There is no further information that may have a bearing on the Project's eligibility, the net GHG emission reductions or removals, or the quantification of the Project's net GHG emission reductions or removals.

2 SAFEGUARDS

2.1 No Net Harm

The Ministry of Environment and Urbanization exempted the project from EIA on 28.01.2021. This decision is given in accordance with the scale of the project, and this decision does not exempt the project from all regulations and legislations. Moreover, the project identification document was prepared and approved by the Ministry. Considering that, it is possible to claim that there is no net harm related to the project.

2.2 Local Stakeholder Consultation

A stakeholder meeting was conducted on 15/11/2022 for Yılmazlar BPP Project. The meeting location was chosen as the Kayı Farm in Derbent, where the plant is located in. The meeting was held in the farm's meeting room, which the local people could easily access. The announcement of the stakeholder meeting was sent by post on 02/11/2022 to the Kutahya Provincial Directorate of Environment and Urbanization, Tavşanlı Municipality as well as NGOs related with environmental issues by mail. Subsequently, village mukhtars were also contacted by telephone to provide further information about the meeting. There was participation in the meeting from the local residents, the deputy Mayor of Tavşanlı Municipality, employees of the Plant, the Project Proponent and the mukhtars of the villages. During the Stakeholder Consultation Meeting, the participants were given detailed information about the planned activity through a visual presentation.

The mechanism for ongoing communication was established with the support of the Mukhtar for stakeholders to forward their requests or concerns regarding the project activities. A logbook for receiving comments from the locals was left in Mukhtar's office. In Türkiye, mukhtars are the most relevant persons in the villages for establishing a grievance mechanism. Their offices are some of the few places the locals visit very often for different purposes, such as requesting some official letters. Whenever a problem arises in the village, as the first thing locals go to the mukhtars office to express their concerns to find solutions and reflect on their issues with the Mukhtar. It has been decided that the Mukhtar will also take an active role in monitoring the participants' requests so that a grievance mechanism situating the Mukhtar at the centre was established for this Project. Mukhtar was provided with all the contact information of the responsible persons for the Project. Whenever a complaint arises from the local people or Mukhtar himself is concerned about the Project, he can directly contact the relevant responsible persons.

In general, local stakeholders were supportive of the Project. Most of the local stakeholders thought the Project would help improve the livelihood of local people and promote local economic development without much adverse environmental impact. Therefore, the project

implementation is regarded as beneficial by most of the stakeholders. No negative comments from the stakeholders were received during this meeting.

2.3 Environmental Impact

Within the scope of project identification document (PID) submitted to Ministry of Environment, Urbanization and Climate Change key environmental impacts were assessed.

Air Pollution

In accordance with the approved PID, emission from biogas utilisation is measured and reported to the Ministry. The Project Proponent committed that emissions are not exceed the threshold given in "Industrial Air Pollution Control Regulation". There is no odor problem expected from digested product since CH₄ that causes odor was used in gas engines. The odor can be expected only in the waste material receiving section of the facility. In order to minimize the odor in the incoming waste, it is ensured that the waste reception units are completely closed.

Water Pollution

Most liquid fertiliser obtained from the biogas plant is recycled back into the system. The remaining liquid fertiliser product formed in the Biogas Plant is supplied to villagers as liquid fertiliser. At every stage of the Project, the provisions of the "Water Pollution Control Regulation", which entered into force after being published in the Official Gazette dated 31.12.2004 and numbered 25687, is complied with. Domestic wastewater generated by the plant is vacuumed by local municipality.

Soil Pollution and Solid Waste

In order not to cause soil pollution, production and final product storage activities are on an impermeable concrete floor. The waste storage area is built on an impermeable concrete floor. The Activity Preliminary Information Form is submitted to the Provincial Directorate of Environment, Urbanization and Climate Change, which is made over the Contaminated Site Information System, within 30 days after the facility is operational. In the event of a situation requiring renewal of the notification, it is renewed.

Domestic solid wastes, which is formed from the social activities of the personnel to be employed in the project, is collected in a sealed and impermeable garbage barrel that is not emit odor during the construction and operation phases. Domestic solid wastes, the relevant articles of the Waste Management Regulation, published in the Official Gazette dated 02.04.2015 and numbered 29314, is complied with regarding transportation, storage and final disposal. Domestic solid wastes originating from the workers at the facility are collected by the Tavşanlı Municipality on a daily basis.

Hazardous waste is likely to be generated during the assembly and operation phases within the project area. Oil change and maintenance of vehicles and machines to be used within the scope of the project are done at fuel stations. Contaminated parts, cloths and gloves that may occur

during the construction phase and the hazardous wastes that arise from the administrative office, maintenance - repair units during the activity in the project area are collected separately from each other in the Hazardous waste storage area to be built within the project area, not exceeding 180 days, and are sent to licensed companies.

Noise Pollution

In Article 22 of the Regulation on the Evaluation and Management of Environmental Noise, criteria regarding the level of noise emitted from industrial facilities, workshops, workshops and workplaces have been determined. In this context, clause a of the relevant article states that "The noise level emitted from each industrial facility to the environment cannot exceed the limit values given in Table-4 in Annex-VII of this Regulation". In this table, it is seen that the limit threshold of noise is 70 dB for industries.

The closest settlement to the project Area is Derbent Village and it is 1250 m away from the project site. The noise level of the facility is less than the threshold limits. Therefore, the settlements shall not be affected by noise pollution.

Since this project activity has received the "EIA Not Required" decision dated 09/02/2021, the project is unlikely to have environmental or social negative impacts. During the current monitoring period (29-September-2021 – 31-March-2023), no issue related to air pollution, water pollution, soil pollution or noise pollution was encountered with the Ministry of Environment, Urbanization and Climate Change or with the local stakeholders.

2.4 Public Comments

This Project was open for public comment on the Verra website after being listed on the pipeline and no comment has been received.

2.5 AFOLU-Specific Safeguards

N/A.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The Project applies AMS.III.D.- "Methane Recovery in animal manure management systems" (version 21.0).

Applied tools are listed below:

- "Project and leakage emissions from anaerobic digesters" (version 02.0)

- “Project Emissions from Flaring” (version 04.0);
- “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0);
- “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion” Version 03.0
- “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” Version 03.0
- “Project and leakage emissions from transportation of freight” Version 01.1.0

Moreover, the Project applies AMS.I.D.- “Grid-connected renewable electricity generation” (version 18.0). Sectoral scopes are listed below:

- “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion” (version 03.0)
- “Tool to calculate the emission factor for an electricity system” (version 07.0)
- “Tool to determine the remaining lifetime of equipment” (version 01.0)
- “Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period” Version 03.0.1

3.2 Applicability of Methodology

Demonstration of applicability conditions for the CDM baseline and monitoring methodology AMS-III.D. (version 21.0), and all above-referred CDM methodological tools are included in the tables below:

Methodology applicability	The proposed Project
AMS-III.D., "Methane Recovery in animal manure management systems "version 21.0	
This methodology is only applicable under the following conditions: (a) The livestock population in the farm is managed under confined conditions; (b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g. river or estuaries), otherwise “AMS-III.H Methane recovery in wastewater treatment” shall be applied; (c) The annual average temperature of baseline site where anaerobic	(a) The livestock population of the farms are under confined conditions. Moreover, these conditions are audited by the state since these conditions are mandatory by regulations. (b) Manure or the streams obtained after treatment is not discharged into natural water resources. (c) The annual average temperature of Kütahya, where the project is located, is 21.7 °C.

manure treatment facility is located is higher than 5 °C; (d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m; (e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario.	(d) Before the Project, anaerobic lagoons were used. The depth of the lagoons was more than 1 m. Moreover, the retention time was more than 1 month. (e) In the baseline, anaerobic lagoons were used. Therefore, no flaring and combustion for gainful use take place.
The Project shall satisfy the following conditions: (a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured; (b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared; (c) The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when remove	(a) Residual waste from animal manure management system will be used as fertilizer. Since fertilizers are used aerobically, no methane emission will occur. (b) Produced biogas will be burnt in order to generate electricity. In emergency situations, there is a flare to ensure all biogas is burnt. Therefore, the project complies with this condition. (c) The cattle barn and BPP are in same area, and chicken litter is brought from nearby farms. The manure will be taken into digesters just shortly after it is brought. Therefore, storage time will not exceed 45 days.
Projects that recover methane from landfills shall use "AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure	The project does not include methane from landfill, wastewater treatment, or composting of animal manure, or co-digestion of animal manure and other organic matters. Therefore, none of the mentioned methodologies is applicable.

shall use "AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology "AMS-III.AO Methane recovery through controlled anaerobic digestion".	
Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the Project, it should be taken into account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component.	The electricity produced by the Project will be delivered to the grid.
New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies".	The Project does not include any capacity additions
The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies".	The project is greenfield project, so there is no replaced equipment.
Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the Project.	Expected emission reduction of the project is 35,855 tCO ₂ e/year which is less than the project. Therefore, the project complies with the condition.

the applicability conditions of the tools referred by the applied methodology AMS-III.D. are explained in the below table.

Tool applicability	The proposed Project
TOOL 14: Project and leakage emissions from anaerobic digesters	
The following sources of project emissions are accounted for in this tool: (a) CO ₂ emissions from consumption of	(a) During the operation of anaerobic digester, electricity will be used, so it is applicable

electricity associated with the operation of the anaerobic digester; (b) CO₂ emissions from consumption of fossil fuels associated with the operation of the anaerobic digester; (c) CH₄ emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester); and (d) CH₄ emissions from flaring of biogas.	(b) During the operation of anaerobic digester, fossil fuel might be used by generator, so it is applicable (c) Physical leakage may occur in digester so it will be applicable. (d) CH₄ will be flared so it is applicable
The following sources of leakage emissions are accounted for in this tool: (a) CH₄ and N₂O emission from composting of digestate; (b) CH₄ emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond.	(a) There will be no composting, so it is not applicable (b) Liquid digestate will be stored for a while in anaerobic storage, so it is applicable
Emission sources associated with N₂O emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies referring to this tool.	Emission sources with N₂O is not applicable since N₂O as per AMS-III.D is considered as negligible.

Tool applicability	The proposed Project
TOOL 06: Project emissions from flaring	
This tool is applicable to the flaring of flammable greenhouse gases where: (a) Methane is the component with the highest concentration in the flammable residual gas; and (b) The source of the residual gas is coal mine methane or a gas from a biogenic	(a) Methane has the highest concentration and (b) Source of the residual gas is biogas So it is applicable.

source (e.g. biogas, landfill gas or wastewater treatment gas).	
The tool is not applicable to the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion. In the case of an enclosed flare, there shall be operating specifications provided by the manufacturer of the flare and these shall be followed by the project participant.	There is not auxiliary fuels and residual has have enough flammable gas.

Tool applicability	The proposed Project
TOOL 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	
If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or TOOL05 Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation Version 03.0 4 of 25 (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil	Scenario A has been selected since electricity is purchased from the grid only and there is no existing captive power plant in the site.

fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	
This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	Scenario I is applicable since electricity is supplied to grid only.

Tool applicability	The proposed Project
TOOL 08: "Tool to determine the mass flow of a greenhouse gas in a gaseous stream"	
Typical applications of this tool are methodologies where the flow and composition of residual or flared gases or exhaust gases are measured for the determination of baseline or project emissions.	These gases are measured for calculating the project emissions from using flare. Therefore, this condition is applicable.
Methodologies where CO ₂ is the particular and only gas of interest should continue to adopt material balances as the means of flow determination and may not adopt this tool as material balances are the cost-effective way of monitoring flow of CO ₂ .	CO ₂ is not the particular and only gas of interest. Therefore, this condition is not applicable.
The underlying methodology should	a) The gaseous stream to the flare unit b) CH₄

specify: (a) The gaseous stream the tool should be applied to; (b) For which greenhouse gases the mass flow should be determined; (c) In which time intervals the flow of the gaseous stream should be measured; and (d) Situations where the simplification offered for calculating the molecular mass of the gaseous stream (equations (3) or (17)) is not valid (such as the gaseous stream is predominantly composed of a gas other than N₂).	c) On a minute basis d) The simplification is valid.
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Tool applicability	The proposed Project
TOOL 12: "Project and leakage emissions from transportation of freight"	
This tool is applicable to project activities which involve freight transportation by road and where transportation is not the main project activity. This tool is not applicable to project activities where transportation is the main source of greenhouse gases emissions. This tool does not provide procedures to estimate baseline emissions from road transportation of freight. The tool only provides to determine CO₂ emissions. CH₄ and N₂O emissions are excluded for simplification as they are small compared to CO₂ emissions.	The transportation is not the main project activity. Therefore, this tool can be used to calculate project emissions from freight transportation.
In addition, the tool is applicable for the determination of project or leakage emissions from freight transportation by rail in project activities where transportation is not the main project activity.	The project activity does not include rail transportation. Therefore, this condition is not applicable.

Summary of applicability of tools under AMS-III.D. (version 21.0)	
TOOL 14 – "Project and leakage emissions from anaerobic digesters"	The Project includes the treatment of a biodegradable fraction of animal manure through anaerobic digestion in an anaerobic digester. The

v.2.0	Project meets the applicability condition of the methodological tool.
TOOL 06 – “Project emissions from flaring” v4.0	There is flaring system in the project to be used in emergency. Therefore, the tool is applicable.
TOOL05 - “Tool to calculate baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0)	The tool is applied for calculating emissions from electricity consumption. For the Project energy consumption is achieved from the grid only and no captive power plant(s) is/are installed at the site of electricity consumption. Hence, the tool is applicable.
TOOL03 - “Tool to calculate project or leakage CO2 emissions from fossil fuel combustion” v03.0	This tool is not used because the project emissions results from the auxiliary equipment use will be negligible compared to the project emissions from electricity consumption or flaring.
TOOL08 – “Tool to determine the mass flow of a greenhouse gas in a gaseous stream” (version 03.0)	This tool is used to calculate the project emissions from using flare unit.
TOOL12 – “Project and leakage emissions from transportation of freight” (version 01.1.0)	This tool is used to calculate the project emissions from the transportation of manure.

Demonstration of applicability conditions for the CDM baseline and monitoring methodology AMS-I.D. (version 18.0), and all CDM methodological tools at section 3.1 are included in the tables below:

Methodology applicability	The proposed Project
AMS-I.D., "Grid-connected renewable electricity generation "version 18.0	
Install a Greenfield plant	The Project is a Greenfield project involving the setting up of a new power plant at a site where there were no power plants in the pre-project scenario. Therefore, the project implies the condition.
Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:	Proposed project is a biogas power plant project so this condition is not applicable.
If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM Project applies only to the renewable component. If the	The Project doesn't have any non-renewable component. The Project involves electricity generation by utilizing biogas produced from animal manure. Hence, this condition is not applicable.

new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	
Combined heat and power (cogeneration) systems are not eligible under this category.	The Project is not a co-generation system, i.e. the simultaneous generation of heat and electrical energy in one process. It consists in a biogas genset unit that produce electricity. The genset heat is recovered in a heat exchanger that transfer the heat to water that is utilized to maintain the required temperature conditions in the mesophilic reactors.
In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the Project should be lower than 15 MW and should be physically distinct from the existing units.	The Project does not involve the capacity addition of renewable energy generation units at an existing renewable power generation facility since there was no renewable energy generation in the project site prior to Project.
In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated, or replacement power plant/unit shall not exceed the limit of 15 MW.	The Project is not a retrofit, rehabilitation or replacement project.
In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with the procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration, other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	The recovered methane is used for electricity generation for supply to grid. The baseline for the electricity component is in accordance with the procedure prescribed under this methodology. The recovered methane is not used for heat generation or cogeneration.
In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	The Project uses animal manure as a fuel which does not include any dedicated plantations.

The applicability conditions of the tools referred by the applied methodology AMS-I.D. are explained in the below table.

Summary of applicability of tools under AMS-I.D (version 18.0)	
"Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion" Version 03.0	This tool is not used because the project emissions results from the auxiliary equipment use will be negligible compared to the project emissions from electricity consumption or flaring.
"Tool to calculate the emission factor for an electricity system" version 07.0	This tool is not used since the Turkish Ministry of Energy and Natural Resources published the Emission factor applicable for the national grid for wind, solar and other renewable projects.
"Tool to determine the remaining lifetime of equipment" Version 01	Project technical lifetime is determined through "Tool to determine the remaining lifetime of equipment" (Version 01). The project option is selected as (a) as below; (a) Use manufacturer's information on the technical lifetime of equipment and compare to the date of first commissioning; The Project is designed for an operational lifetime of 30 years.
"Assessment of the validity of the original/current baseline and update of the baseline at the renewal of the crediting period" Version 03.0.1	This tool is not used due to the fact that the Project is not at the renewal of the crediting period.

3.3 Project Boundary

As noted above, the baseline and monitoring methodologies applied to the proposed Project are AMS-III.D. "Methane recovery in animal manure management systems", Version 21.0. and AMS-I.D. "Grid-connected renewable electricity generation" Version 18.0.

In accordance with these Methodologies, the project boundary includes:

- The livestock;
- Animal manure management systems (including centralized manure treatment plant);
- Facilities which recover and flare/combust or use methane;
- The project power plant and all power plants connected physically to the electricity system that the proposed project power plant is connected to.

The relevant GHG sources included in or excluded from the project boundary are shown on the Table below.

Source		Gas	Included?	Justification/Explanation
Baseline	Emissions from the manure treatment processes	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted
		CH ₄	Yes	The major source of emissions in the baseline
		N ₂ O	No	Excluded for simplification. This is conservative.
	Emissions from electricity consumption/generation	CO ₂	Yes	Electricity is consumed from the grid in the baseline scenario
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
Project	Physical leakage of biogas	CO ₂	No	Excluded for simplification. This emission source is assumed to be very small.
		CH ₄	Yes	The major source of emissions. Leakage from biogas digester has been considered.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from on-site electricity use	CO ₂	Yes	These emissions are accounted for since the project activity consumes the electricity from the grid.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Emissions from the flaring or combustion of gas stream	CO ₂	No	Flare system will be in use only emergency. If there is any flaring, emission reduction will be excluded during crediting period.
		CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
		CO ₂	Yes	May be an important emission source. Hence, they are accounted

Source	Gas	Included?	Justification/Explanation
Emissions from incremental transportation distances	CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
Emissions from storage of manure	CO ₂	No	Excluded for simplification. This emission source is assumed to be very small.
	CH ₄	Yes	Might be a major emission source.
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.

Schematic demonstration of the project boundary also shown in figure below.

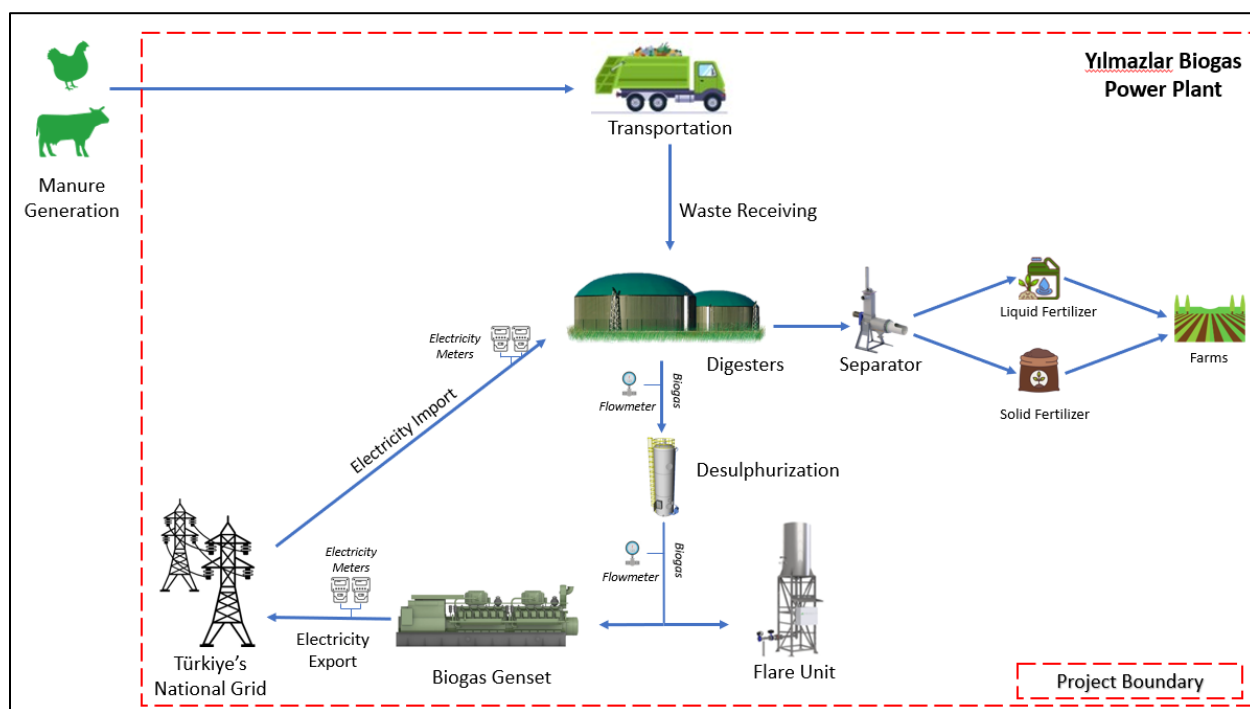


Figure 2 Schematic Demonstration of the Project Boundary

3.4 Baseline Scenario

The proposed Project has two major GHG emission reduction components. The first one is the reduction of methane emissions resulting from improved animal manure management, whilst the second one is the use of biogas recovered from improved animal manure management for

generating power and feeding the electricity generated directly to the national electricity grid network. Assessment of baseline scenario for the both components are provided below:

As per AMS-III.D version 21.0

In line with the applied methodology AMS-III.D. version 21.0, New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the General guidelines for SSC CDM methodologies version 23.1⁷. The Guidelines describes “procedure to determine baseline scenario for Type II and Type III Greenfield projects” which is a 4-step process to be followed in order to determine whether the requirements are met⁸. Steps to be followed is listed below.

Step 1

Identify the various alternatives available to the project proponent that deliver comparable levels of service, including the proposed project activity or PoA undertaken without being registered as a CDM project activity or PoA.

Step 2

List the alternatives identified in Step 1 that are in compliance with local regulations. If any of the identified baselines is not in compliance with local regulations, then exclude that alternative from further consideration).

Step 3

Eliminate and rank the alternatives identified in Step 2 taking into account barrier tests specified in the “Guidelines on the demonstration of additionality of small-scale project activities”.

Step 4

The project activity or PoA is eligible under the methodology if only one alternative remains that:

(a) Is not the proposed project activity or PoA without being undertaken as registered CDM project activity or PoA; and

(b) Corresponds to one of the baseline scenarios provided in the methodology.

If more than one alternative remains that correspond to a baseline scenario provided in the methodology, choose the alternative with the lowest emissions as the baseline

⁷ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210211212225226/MethSSC_Guid25ver23.1.pdf

⁸ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210211212225226/MethSSC_Guid25ver23.1.pdf

Step 1 – Identify the various alternatives to the project proponent that deliver comparable levels of service, including the proposed project activity undertaken without being registered as a CDM project activity.

The project activity is a greenfield project. According to the applied methodology for greenfield projects, the baseline scenario is selected from the complete set of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories (Volume 4, Chapter 10, Table 10.14)⁹.

Manure management systems are listed below:

- Pasture, Range and Paddock
- Solid Storage
- Liquid/Slurry Storage
- Anaerobic Lagoons
- Pit Storage
- Dry Lot
- Daily Spread
- Anaerobic Digestion – Biogas
- Burned for Fuel

Step 2 - List the alternatives identified in Step 1 that are in compliance with local regulations. If any of the identified baselines is not in compliance with local regulations, then exclude that alternative from further consideration).

Information for the alternatives are given in the table below.

IPCC Alternatives	Justification
Pasture, Range and Paddock	In accordance with “Regulation on Control of Odor Generating Emissions”, collected manure shall be stored before it is spread to croplands. Therefore, the alternative is not applicable.
Solid Storage	Solid storage system is not in line with current “Regulation on Livestock Farms” entered into force in 2006.
Liquid/Slurry Storage	This alternative does not comply with the local regulations since “Air Quality Regulation of Türkiye” states that liquid wastes shall be stored in

⁹ https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/4_Volume4/V4_10_Ch10_Livestock.pdf

	impervious storage areas and transferred with closed containers.
Anaerobic Lagoons	In accordance with “Regulation on Livestock Farms” entered into force in 2006, animal manure shall be kept in lagoons made in concrete so that it shall be leakage-proof. Therefore, the alternative complies with the local regulations.
Pit Storage	Pit storage system was compliant with old regulations, but it is not allowed anymore. Animal farms with pit storage that was implemented before the new regulation, are asked to renew their storage system.
Dry Lot	In accordance with “Regulation on Control of Odor Generating Emissions”, collected manure shall be stored before it is spread to croplands. Therefore, the alternative is not applicable.
Daily Spread	In accordance with “Application Communique on Animal By-Products Regulation Not Used for Human Consumption”, collected manure shall be stored before it is spread to croplands. Therefore, the alternative is not applicable.
Anaerobic Digestion - Biogas	Anaerobic digestion with biogas would associate with high investment costs and skilled manpower. There is also no legal requirement for aerobic treatment.
Burned for Fuel	Burning animal manure is not allowed in accordance with local regulations.

Step 3: Eliminate and rank the alternatives identified in Step 2 taking into account barrier tests specified in the “Guidelines on the demonstration of additionality of small-scale project activities”.

Alternative Scenario	Justification
The manure is disposed in an uncovered anaerobic lagoon;	This alternative is considered a realistic alternative since there is no regulation that restricts the scenario and applicable. The

	manure is disposed in an uncovered anaerobic lagoon. This is also the real baseline scenario.
Anaerobic Digestion - Biogas	This alternative is proposed project activity. Although there are no regulatory restrictions to implement this scenario, the alternative is eliminated due to technological and investment barriers.

Step 4: If only one alternative remains that is not the proposed project activity undertaken without being registered as a CDM project activity and also corresponds to one of the baseline scenarios provided in the methodology, then the project activity is eligible under the methodology. If more than one alternative remains that correspond to a baseline scenario provided in the methodology, choose the alternative with the lowest emissions as the baseline.

As a result of steps above, only one alternative scenario maintains all the requirements as baseline scenario. That is why, the proposed project activity shall be eligible under AMS-III.D. methodology.

As per AMS-I.D version 18.0

In accordance with paragraph 19 of AMS-I.D. v.18, *“The baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.”*

Since the project will generate renewable energy by utilization of biogas and supply it to national grid which is mainly driven by fossil fuel fired power plants, it will reduce CO₂ emissions that would have been generated without the project activity.

3.5 Additionality

Section 3.13.1. of VCS Standard, v4.1, clearly mandates that *“Additionality shall be demonstrated and assessed in accordance with the requirements set out in the methodology applied to the project”*. Also, same section of paragraph 2 of the VCS Standard V4.1 states that *“the project proponent shall demonstrate to the validation/verification body that the simplified procedure is appropriate to apply to the project considering the project characteristics, including the context in which the project activity takes place.”*

The paragraph 15 of the approved methodology of AMS-III.D., version 21.0, applied to the proposed project activity, states that *“Project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the ‘Guidelines on the demonstration of additionality of small-scale project activities’”*.

As noted in the section 3.4, there is no legal regulation in Türkiye that requires the collection and destruction of methane from livestock manure. Thus, it is not required to apply the “*Guidelines on the demonstration of additionality of small-scale project activities*”. Accordingly, Type III component of the proposed project activity, as methane recovery from animal manure, shall be considered as deemed additional.

Moreover, the paragraph 16 of AMS-III.D., version 21.0. clarifies that “*This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component*”.

Within the boundaries of the Type III component of the proposed project the project captures and utilizes biogas from animal manure in anaerobic digesters to generate electricity with biogas engines/units with the total installed capacity that is lower than 5 MW. Hence, as per the paragraph 16 of AMS-III.D., version 21.0, it is not required to apply the ‘*Guidelines on the demonstration of additionality of small-scale project activities*’ to the Type I component of the proposed Project which is grid-connected renewable electricity generation. Accordingly, Type I component of the proposed Project, as grid-connected renewable electricity generation, shall also be considered as deemed additional.

Therefore, the proposed Project, as per the approved methodology of AMS-III.D., version 21.0, is deemed to be additional without further documents of barriers as per “*Guidelines on the demonstration of additionality of small-scale project activities*”.

3.6 Methodology Deviations

As per the applied methodology AMS-III.D, v21.0, to estimate W_{site} monitoring parameter, when IPCC values of VS are adjusted for site specific animal weight, sampling procedures shall be used. Since there is only one cattle farm with the animal number of 2500, all animals are weighted at the farm yearly and no sampling approach is needed for the dairy cattle farm. However, for the chicken broilers farms, the baseline surveys had to used to estimate W_{site} parameter because owners of chicken farms do not allow the project owner to enter the farms since chickens are very susceptible to diseases. The baseline surveys were signed by the owners of the farms and these surveys have been provided to the VVB at the time of this joint validation/verification process.

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

The facility has commissioned as of 27 June 2021 and officially started its operation. However, project was not fully functional until September 2021 since test productions of biogas etc. was held on. Apart from that, during the operation phase of the project, there is no major event that impacts the GHG emission reductions occurred. Milestones of the project are given in the table below:

Date	Action
01-December-2020	Installation of the Flare Equipment
09-February-2021	EIA Exemption Letter
20-May-2021	EMRA Generation License
09-June-2021	Grid Connection Agreement
27-June- 2021	Yilmazlar Biogas Power Plant Provisional Acceptance
29-September- 2021	Start Date of First Crediting and Monitoring Period (i.e. start date of the biogas production with the anaerobic digesters and electricity generation) (Start Date of Manure Collection in Yilmazlar Biogas Power Plant)
15-November-2022	Local Stakeholder Consultation Meeting

5 ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emission calculations according to AMS.III.D (version 21.0):

Before the Project, the manure would decay anaerobically. Therefore, option (a) presented in AMS-III.D., version 21.0 paragraph 17 is utilised. Since there are both cattle and poultry farms that generates manure, emissions from both farms are considered separately and added into calculation.

In accordance with the option(b) baseline emissions are determined based on directly measured quantity of manure and its specific volatile solids content, as follows:

$$BE_y = GWP_{CH_4} \times D_{CH_4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{BI,j}$$

Equation 1

Where;

BE_y	=	Baseline emissions in year y (t CO ₂ e)
GWP_{CH_4}	=	Global Warming Potential (GWP) of CH ₄ applicable to the crediting period (t CO ₂ e/t CH ₄)
D_{CH_4}	=	CH ₄ density (0.00067 t/m ³ at room temperature (20 °C) and 1 atm pressure)
LT	=	Index for all types of livestock
j	=	Index for animal manure management system
MCF_j	=	Annual methane conversion factor (MCF) for the baseline animal manure management system j
$B_{0,LT}$	=	Maximum methane producing potential of the volatile solid generated for animal type LT (m ³ CH ₄ /kg-dm)
$N_{LT,y}$	=	Annual average number of animals of type LT in year y (numbers)
$VS_{LT,y}$	=	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)
$MS\%_{BI,j}$	=	Fraction of manure handled in baseline animal manure management system j
UF_b	=	Model correction factor to account for model uncertainties (0.94)

Yılmazlar biogas power plant receives waste from 12 different farms. List of the farms and information related to farms are given in the table below:

Name of the Farm	Type of animal	Number of Animals	Average Weight of Animal (kg)	Rate of received manure
Kayı Farm	Dairy Cattle	2,500	465	100%
As Tarım	Chicken Layer	50,000	1.5	50%
Baş Tarım	Chicken Layer	220,000	1.5	50%
Özkantar Gıda	Chicken Layer	95,000	1.5	50%
Halil Yeşil Yumurta Üretim	Chicken Layer	12,600	1.5	50%
Yoncalı Gıda	Chicken Layer	350,000	1.5	50%
Yumurtalar Tarım	Chicken Layer	28,000	1.5	50%
Emre Kayner	Chicken Layer	18,500	1.5	50%
Halil Küçük	Chicken Layer	15,000	1.5	50%
Saydam Tarım	Chicken Layer	22,000	1.5	50%
Bayraktar	Chicken Layer	330,000	1.5	50%
Ali Aydın	Chicken Layer	70,000	1.5	100%

To calculate $N_{LT,y}$, the following formula is used:

$$N_{LT,y} = N_{da,y} \times \frac{N_{p,y}}{365}$$

Equation 2

Where:

$N_{da,y}$ = Number of days animal is alive in the farm in the year y (numbers)

$N_{p,y}$ = Number of animals produced annually of type LT for the year y (numbers)

$N_{da,y}$ is considered as 365 values as per the plant logs which were obtained from the local farmers. Also, again as per the plant logs, $N_{p,y}$ values are taken as follows:

$N_{p,y}$ for dairy cattle = 2500 ; $N_{p,y}$ for chicken broiler = 1,211,100

Therefore, $N_{LT,y}$ values are 2500 for dairy cattle and 1,211,100 for chicken broiler.

To calculate $VS_{LT,y}$, the following formula is used:

$$VS_{LT,y} = \left(\frac{W_{site}}{W_{default}} \right) \times VS_{default} \times nd_y$$

Equation 3

Where

W_{site} = Average animal weight of a defined livestock population at the project site (kg)

$W_{default}$ = Default average animal weight of a defined population, this data is sourced from IPCC 2006 (kg)

$VS_{default}$ = Default value for the volatile solid excretion rate per day on a dry-matter basis for a defined livestock population (kg-dm/animal/day)

nd_y = Number of days treatment plant was operational in year y

W_{site} values are taken from the plant logs which were obtained from the local farmers. As per the logs, the values are 465 kg for dairy cattle and 1.5 kg for chicken broiler.

$W_{default}$ values are taken from IPCC Table 10A.5. The relevant values are 275 kg for dairy cattle and 0.7 kg for chicken broiler.

$VS_{default}$ values are taken from IPCC Table 10.13A. The relevant values are 10.7 kg-VS/(1000 kg animal mass)/day for dairy cattle and 17.7 kg-VS/(1000 kg animal mass)/day for chicken broiler.

nd_y is taken as 365 days as per project proponent's declaration.

$$BE_{Kayı \text{ Farm},y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.24 \times 2500 \times 1816.0575 \times 1) = 14,027 \text{ tCO}_2\text{e/y}$$

$$BE_{As \text{ Tarım},y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 50000 \times 7.7745 \times 0.5) = 976 \text{ tCO}_2\text{e/y}$$

$$BE_{Baş \text{ Tarım},y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 220000 \times 7.7745 \times 0.5) = 4,294 \text{ tCO}_2\text{e/y}$$

$$BE_{Özkantar ,y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 95000 \times 7.7745 \times 0.5) = 1,854 \text{ tCO}_2\text{e/y}$$

$$BE_{Halil \text{ Yeşil },y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 12600 \times 7.7745 \times 0.5) = 246 \text{ tCO}_2\text{e/y}$$

$$BE_{Yoncalı ,y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 350000 \times 7.7745 \times 0.5) = 6,831 \text{ tCO}_2\text{e/y}$$

$$BE_{Yumurtaları ,y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 28000 \times 7.7745 \times 0.5) = 546 \text{ tCO}_2\text{e/y}$$

$$BE_{Emre \text{ Kayneri },y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 18500 \times 7.7745 \times 0.5) = 361 \text{ tCO}_2\text{e/y}$$

$$BE_{Halil \text{ Küçükü },y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 15000 \times 7.7745 \times 0.5) = 293 \text{ tCO}_2\text{e/y}$$

$$BE_{Saydamı ,y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 22000 \times 7.7745 \times 0.5) = 429 \text{ tCO}_2\text{e/y}$$

$$BE_{\text{Bayraktar},y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 330000 \times 7.7745 \times 0.5) = 6,440 \text{ tCO}_2\text{e/y}$$

$$BE_{\text{Ali Aydın},y} = 28 \times 0,00067 \times 0,94 \times (0.73 \times 0.39 \times 70000 \times 7.7745 \times 1) = 2,732 \text{ tCO}_2\text{e/y}$$

As a result, **BE_y = 39,028 tCO₂e/y**

Baseline emission calculations according to AMS.I.D (version 18.0):

In accordance with this methodology, baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the Project. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{\text{grid},y}$$

Equation 4

Where;

BE_y = Baseline emissions in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM Project in year y (MWh)

$EF_{\text{grid},y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

The project is greenfield project so, in accordance with the paragraph 26 of AMS-I.D., the equation below shall be used for calculation of $EG_{PJ,y}$:

$$EG_{PJ,y} = EG_{PJ,\text{facility},y}$$

Equation 5

$EG_{PJ,\text{facility},y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

As per the applied methodology, the emission factor for the electricity generation $EF_{\text{grid,CM},y}$ is determined according to “Tool to calculate the emission factor for an electricity system”, version 07.0. According to the Tool, the “combined margin” emission factor is calculated through the following six steps as per the “Tool to calculate the emission factor for an electricity system”, version 07.0:

Step (1): Identify the relevant electricity system;

Step (2): Choose whether to include off-grid power plant in the project electricity system (optional);

Step (3): Select a method to determine the operating margin (OM);

Step (4): Calculate the operating margin emission factor according to the selected method;

Step (5): Calculate the build margin (BM) emission factor;

Step (6): Calculate the combined margin (CM) emission factor

Step 1: Identify the relevant electricity system

According to the applicability of “Tool to calculate the emission factor for an electricity system”, version 07.0, the relevant project electricity system should be identified by the project owners. The chosen electric power system is the Turkish National Grid, which is physically connected through transmission and distribution lines to the project activity. National grid is dominated by fossil fuels such as coal, diesel and natural gas.

Step 2: Choose whether to include off-grid power plants in the project electricity system

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, project owners may choose between the following two options to calculate the operating margin and build margin emission factor:

- Option I: Only grid power plants are included in the calculation
- Option II: Both grid power plants and off-grid power plants are included in the calculation

The project participants choose Option I and therefore only grid power plants are included in the calculation.

Step 3: Select a method to determine the operating margin (OM)

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods:

- a) Simple OM; or
- b) Simple adjusted OM; or
- c) Dispatch data analysis OM; or
- d) Average OM.

The simple OM method has been selected for calculation.

Step 4: Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units. The simple OM may be calculated by one of the following two options:

Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit;
or

Option B: Based on the total net electricity generation of all power plants serving and the fuel types and total fuel consumption of the project electricity system.

Ministry of Energy and Natural resources of Turkey has published the data for OM using simple OM calculation following the option A.

$$EF_{grid,OM,y} = 0.7424 \text{ tCO}_2\text{e/MWh}$$

For this approach (simple OM) to calculate the operating margin, the subscript m refers to the power plants/units delivering electricity to the grid, not including low-cost/must-run power plants/units. In addition, project owners has chosen to apply ex-post option for OM emission factor as described in the paragraph 42(b) of Tool 07, v07.0

Step 5: Calculate the build margin (BM) emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

Option 1: for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PDMR submission to the VVB for the project verification.

Option 2: for the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available.

For the proposed project activity, the project participants choose Option 1 in terms of availability of vintage of data. Based on 2020 data, Ministry of Energy and Natural resources of Turkey has calculated build margin and published the data:

$$EF_{grid,OM,y} = 0.3680 \text{ tCO}_2\text{e/MWh}$$

Step 6: Calculate the combined margin emissions factor

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Equation 6

Where:

$EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (t CO₂/MWh)

$EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (t CO₂/MWh)

w_{OM} = Weighting of operating margin emissions factor (per cent)

w_{BM} = Weighting of build margin emissions factor (per cent)

According to CDM Tool 07 “Tool to calculate emission factor for an electricity system” v07.0¹⁰, paragraph 86(b) w_{OM} and w_{BM} for the first crediting period are given below:

$$w_{OM} = 0.5$$

$$w_{BM} = 0.5$$

As a result:

$$EF_{grid,CM,y} = 0.7424 \times 0.5 + 0.3680 \times 0.5 = 0.5552 \text{ tCO}_2/\text{MWh}$$

So annual estimated emission reduction,

$$BE_y = EG_{pj,y} * EF_{grid,CM,y} = 11,700 \times 0.5552 = 6,495 \text{ tCO}_2/\text{year}$$

5.2 Project Emissions

Project Emissions as per AMS-III.D.

The project emissions in year y are calculated for each alternative waste treatment process implemented in the Project as follows:

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} \quad \text{Equation 7}$$

Where:

PE_y = Project emissions in year y (t CO₂e)

$PE_{PL,y}$ = Emissions due to physical leakage of biogas in year y (t CO₂e)

¹⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

$PE_{flare,y}$	=	Emissions from flaring or combustion of the biogas stream in the year y (t CO ₂ e)
$PE_{power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO ₂ e)
$PE_{transport,y}$	=	Emissions from incremental transportation in the year y (t CO ₂ e)
$PE_{storage,y}$	=	Emissions from the storage of manure (t CO ₂ e)

Project emissions from anaerobic digestion ($PE_{PL,y}$)

$PE_{PL,y}$ is calculated according to “TOOL14: Project and leakage emissions from anaerobic digesters” as per AMS-III.D paragraph 21(b) stated.

As per paragraph 23 of the “TOOL14: Project and leakage emissions from anaerobic digesters”, The project emission associated with the anaerobic digester ($PE_{PL,y}$) equals to ($PE_{CH_4,y}$) and is determined as follows:

$$PE_{CH_4,y} = Q_{CH_4,y} \times EF_{CH_4,default} \times GWP_{CH_4}$$

Equation 8

Where:

$PE_{CH_4,y}$	=	Project emissions associated with the anaerobic digester in year y (t CO ₂ e)
$Q_{CH_4,y}$	=	Quantity of methane produced in the anaerobic digester in year y (t CH ₄)
$EF_{CH_4,default}$	=	Project emissions from fossil fuel consumption associated with the anaerobic digester in year y (t CO ₂ e)
GWP_{CH_4}	=	Project emissions of methane from the anaerobic digester in year y (t CO ₂ e)

And $Q_{CH_4,y}$ is calculated as follows:

$$Q_{CH_4,y} = Q_{biogas,y} \times f_{CH_4,default} \times \rho_{CH_4}$$

Equation 9

$Q_{CH_4,y}$	=	Quantity of methane produced in the digester in year y (t CH ₄)
$Q_{biogas,y}$	=	Amount of biogas collected at the digester outlet in year y (Nm ³ biogas)
$f_{CH_4,default}$	=	Default value for the fraction of methane in the biogas (m ³ CH ₄ / m ³ biogas)

ρ_{CH_4} = Density of methane at normal conditions ($t\ CH_4 / Nm^3\ CH_4$)

Parameter	Value
$EF_{CH_4, default}$	0.05
GWP_{CH_4}	28
$f_{CH_4, default}$	0.6
ρ_{CH_4}	0.00067 t/m^3

Determination of project emissions from flaring of biogas ($PE_{flare,y}$)

In accordance with paragraph 22 of AMS-III.D, “In the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool “Project emissions from flaring”. If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. However, this use of the recovered biogas shall be included in the project boundary and its output shall be monitored in order to ensure that the recovered biogas is actually destroyed, even if the emission reductions from this component are not claimed.”

Since the methane is used for energy production in biogas engines, destruction efficiency is taken as 100%.

However, there is also flare unit in the plant which has open flare. The flare is installed for the case of emergency and malfunction in the plant. Thus, $PE_{flare,y}$ is considered as zero. Nevertheless, the monitoring plan of the facility shall measure the quantity of captured methane fed to the flare. If such measurement is positive, the emissions from flaring the biogas shall be determined following the procedure described in the methodological tool project emissions from flaring, version 04.0. To be conservative, efficiency of flaring in flare unit is taken as 0%.

Project emissions from flaring are calculated as the sum of emissions from each minute m , based on the methane mass flow in the residual gas ($F_{CH_4, RG, m}$) and the flare efficiency

($\eta_{flare, m}$), as follows:

$$PE_{flare,y} = GWP_{CH_4} \times \sum_{m=1}^{525600} F_{CH_4, RG, m} \times (1 - \eta_{flare, m}) \times 10^{-3}$$

Equation 10

Where:

$PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO_2e)

$F_{CH_4, RG, m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{\text{flare},m}$ = Flare efficiency in minute m

Parameter	Value
$\eta_{\text{flare},m}$, biogas engine	100%
$\eta_{\text{flare},m}$, flare unit	0%

To calculate $F_{\text{CH}_4,\text{RG},m}$, Tool 08 (version 03.0) is used. Option 2: “Simplified calculation without measurement of the moisture content”, Option A is chosen.

Flow measurement on a dry basis is not doable for a wet gaseous stream. Therefore, it is necessary to demonstrate that the gaseous stream is dry to use this option. There are two ways to do this:

- Measure the moisture content of the gaseous stream ($\text{CH}_2\text{O},t,\text{db},n$) and demonstrate that this is less or equal to 0.05 kg $\text{H}_2\text{O}/\text{m}^3$ dry gas; or
- Demonstrate that the temperature of the gaseous stream (T_t) is less than 60°C (333.15 K) at the flow measurement point.

As per (b), the temperature of the gaseous stream (T_t) is less than 60°C (333.15 K) according to the thermocouple data.

$$F_{i,t} = V_{t,\text{db}} \times v_{j,t,\text{db}} \times \rho_{i,t}$$

with

$$\rho_{i,t} = \frac{P_n \times MM_i}{R_u \times T_n}$$

- $F_{i,t}$ = Mass flow of greenhouse gas i in the gaseous stream in time interval t (kg gas/h)
- $V_{t,\text{db}}$ = Volumetric flow of the gaseous stream in time interval t on a dry basis (m^3 dry gas/h)
- $v_{i,t,\text{db}}$ = Volumetric fraction of greenhouse gas i in the gaseous stream in a time interval t on a dry basis (m^3 gas i/ m^3 dry gas)
- $\rho_{i,t}$ = Density of greenhouse gas i in the gaseous stream in time interval t (kg gas i/ m^3 gas i)
- P_n = Absolute pressure at normal conditions (Pa)
- MM_i = Molecular mass of greenhouse gas i (kg/kmol)
- R_u = Universal ideal gases constant (Pa.m³/kmol.K)

T_n = Molecular mass of the gaseous stream in a time interval t on a dry basis (kg dry gas/kmol dry gas)

Determination of project emissions from electricity consumption ($PE_{EC,y}$)

In accordance with paragraph 23 of AMS-III.D, “*Project emissions from electricity and fossil fuel consumption are determined by following the methodological tool “Project and leakage emissions from anaerobic digesters”.*”

Option 1 is chosen to determine project emissions from electricity consumption. As per Option 1 of Tool 14, $PE_{EC,y}$ shall be calculated using the “Tool 05 Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” version 03.0, where the project emission source j referred to in the tool is the total electricity consumption associated with the anaerobic digestion facility.

According to Paragraph 16 of Tool 05, project emissions from the consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y}) \quad \text{Equation 11}$$

Where:

- $PE_{EC,y}$ = Project emissions from electricity consumption in year y (t CO₂ / yr)
- $EC_{PJ,j,y}$ = Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)
- $EF_{EF,j,y}$ = Emission factor for electricity generation for source j in year y (t CO₂/MWh)
- $TDL_{j,y}$ = Average technical transmission and distribution losses for providing electricity to source j in year y
- j = Sources of electricity consumption in the project

However, since the Project involves electricity generation, it is expected that the plant will generate more electricity than it consumes. Moreover, net electricity generation and emission reduction due to renewable energy generation will be claimed under AMS-I.D. v18.0 and energy consumption will be subtracted from energy generated and EPIAS data has already shows transmission and distribution losses for the electricity generated to grid and consumed from the grid. Therefore, PE_{power} will be considered only if there is energy consumption without energy generation in monthly basis.

Determination of the emission factor for electricity generation $EF_{EF,j,y}$

According to the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0), for the project in the case of Scenario A:

Electricity consumption from the grid, the project participants choose Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the “Tool to calculate the emission factor for an electricity system” (version 07.0)

For the emission factor, the Turkish Ministry of Energy and Natural Resources publication, which indicates Türkiye's National Electric Grid Emission Factor for the year 2020, was used. The publication includes calculated Emission Factor values that are Operating Margin (OM), Growth Based Margin (Build Margin-BM) and Combined Margin (CM) Emission Factors for the relevant year with the usage of the IPCC's Clean Development Methodology Tool 07-v07.0. For this calculation, information regarding the data set is given below in detail;

- TEİAŞ Türkiye's electricity generation-consumption and loss statistics,
- Common prepared report under Türkiye's National Greenhouse Gas Inventory Reporting Format. - Common Reporting Format (CRF) tables for electricity generation (1.A.1.a.i) emission values
- Chronological order of power generation plants from TEİAŞ Load Dispatch Department with commissioning dates, plant names, fuel types, installed power values, electricity generation for the calculated year
- Checking off Volunteers from the websites of Gold Standard (GS), Verified Carbon Standard (VCS) and Global Carbon Council for the ownership status of the carbon reduction certificate and,
- From Clean Development Mechanism (CDM) Tool 009- V03.0, Power plant efficiency figures are used

According to this publication;

- Operating Margin-OM; 0.7424 tCO₂/MWh
- Build Margin-BM; 0.3680 tCO₂/MWh
- Combined Margin-CM (other renewables than solar and wind); 0.5552 tCO₂/MWh¹¹

Determination of project emissions from incremental transportation in the year (PE_{transport})

In accordance with AMS-III.D., version 21.0., PE_{transp,y} is calculated as per the paragraph 14 of AMS-III.AO, version 01.0., which states project emissions due to incremental transport distances (PE_{transp,y}) are calculated based on the incremental distances between:

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<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/EmisyonFaktorleri/BilgiFormu.pdf>

$$PE_{transp,y} = (Q_y / CT_y) * DAF_w * EF_{CO2,transport} + (Q_{res-waste,y} / CT_{res-waste,y}) * DAF_{res-waste} * EF_{CO2,transport} \quad \text{Equation 12}$$

Where:

- Q_y = Quantity of raw waste/manure treated and/or wastewater co-digested in the year y (tonnes)
- CT_y = Average truck capacity for transportation (tonnes/truck)
- DAF_w = Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck)
- $EF_{CO2,transport}$ = CO_2 emission factor from fuel use due to transportation (kg CO_2 /km, IPCC default values or local values may be used)
- $Q_{res-waste,y}$ = Quantity of residual waste produced in year y (tonnes)
- $CT_{res-waste,y}$ = Average truck capacity for residual waste transportation (tonnes/truck)
- $DAF_{res-waste}$ = Average distance for residual waste transportation (km/truck)

The distances between farms and Yılmazlar BPP is given in the table below.

Name of the Farm	Distance to the plant (km)
Kayı Farm	0
As Tarım	18
Baş Tarım	25
Özkantar Gıda	25
Halil Yeşil Yumurta Üretim	25
Yoncalı Gıda	23
Yumurtalar Tarım	22
Emre Kayner	24
Halil Küçük	24
Saydam Tarım	24
Bayraktar	24
Ali Aydın	100

In accordance with studies poultry can defecate 3-4% of its weight in a day¹². So, for the calculation 4% has taken to be conservative and Q_y has found accordingly.

¹² <https://dergipark.org.tr/tr/pub/gbad/issue/29696/319526>

Project determine emissions from the storage of manure

Since the storage time of the manure after removal from the animal barns, including transportation, does not exceed 24 hours before being fed into the anaerobic digester in the proposed Project, $PE_{storage,y}$ equals to zero.

Project Emissions as per AMS-I.D.

As per the paragraph 39 of AMS-I.D., version 18.0., for most renewable energy project activities, $PE_y = 0$. However, for the following categories of project activities, project emissions have to be considered following the procedure described in ACM0002 v21.0:

- (a) Emissions related to the operation of geothermal power plants (e.g. non condensable gases, electricity/fossil fuel consumption);
- (b) Emissions from water reservoirs of hydro power plants.

As per the paragraph 40 of AMS-I.D., CO_2 emissions from on-site consumption of fossil fuels due to the Project shall be calculated using the latest version of the “Tool to calculate project or leakage CO_2 emissions from fossil fuel combustion”. Also, the paragraph 41 of AMS-I.D. states that in case biomass is sourced from dedicated plantations, the procedures in the tool “Project emissions from cultivation of biomass” shall be used

Since none of the exception stated in the relevant paragraphs of AMS-I.D. mentioned above is applicable to the proposed Project, PE_y as per AMS-I.D., version 18.0., shall be accounted as 0.

5.3 Leakage

In accordance with AMS-III.D. the leakage emission of the project is calculated following the relevant procedure in the methodological tool “Project and leakage emissions from anaerobic digesters”, version 02.0 and formula in the tool is given below:

$$LE_{AD,y} = LE_{storage,y} + LE_{comp,y}$$

Equation 13

Where:

$LE_{AD,y}$ = Project emissions from electricity consumption in year y ($t\ CO_2 / yr$)

$LE_{storage,y}$ = Leakage emissions associated with storage of digestate in year y ($t\ CO_2e$)

$LE_{comp,y}$ = Leakage emissions associated with composting digestate in year y ($t\ CO_2e$)

Since solid digestate is not disposed in a SWDS or a stockpile, $LE_{storage,y}$ are not considered for the solid digestate part. The solid digestate is distributed directly to the local farmers for free. Therefore, $LE_{AD,y}$ equals to $LE_{storage,y}$ from liquid digestate. Leakage emissions for is calculated as per Tool 14 “Project and leakage emissions from anaerobic digesters” version 02.0. To calculate leakage emission from liquid digestate, Option 2 has chosen which presented in same tool,

heading 6.2.1.3. According to the tool, leakage emission from liquid digestate is calculated by the following formula.

$$LE_{storage,y} = F_{ww,CH4,default} \times Q_{CH4,y} \times GWP_{CH4}$$

Equation 14

Where:

$LE_{storage,y}$	=	Leakage emissions associated with storage of digestate in year y (t CO ₂ e)
$F_{ww,CH4,default}$	=	Default factor representing the remaining methane production capacity of liquid digestate (fraction)
$Q_{CH4,y}$	=	Quantity of methane produced in the digester in year y (tCH ₄)
GWP_{CH4}	=	Global warming potential of CH ₄ (tCO ₂ / tCH ₄)

As per AMS-I.D. version 18.0, “General guidance on leakage in biomass project activities shall be followed to quantify leakages pertaining to the use of biomass residues.”. Since this is not applicable for proposed project, leakage emission resulted from AMS-I.D. v.18 equals to zero.

5.4 Estimated Net GHG Emission Reductions and Removals

Net GHG Emission Reduction and Removals as per AMS-III.D

To calculate the emission reductions as per AMS-III.D, following equation shall be applied as per paragraph 27 of the methodology:

$$ER_{y,ex\ post} = \min[(BE_{y,ex\ post} - PE_{y,ex\ post}), (MD_y - PE_{power,y,ex\ post})]$$

Equation 15

Where:

$ER_{y,ex\ post}$	=	Emission reductions achieved by the Project based on monitored values for year y (t CO ₂ e)
$BE_{y,ex\ post}$	=	Baseline emissions calculated using equation 1 (for projects using option in paragraph 17(a)) using ex post monitored values of $N_{LT,y}$ and if applicable $VS_{LT,y}$. For projects using option in paragraph 17(a), the ex post monitored values for $Q_{manure,j,LT,y}$ and $SVS_{j,LT,y}$ are used
$PE_{y,ex\ post}$	=	Project emissions calculated using equation 6 using ex post monitored values of $N_{LT,y}$, $MS\%_{i,y}$, $MS\%_{i,All}$, $Q_{res\ waste,y}$ and if applicable $VS_{LT,y}$

MD_y = Methane captured and destroyed or used gainfully by the Project in year y (t CO₂e)

$PE_{power,y,ex\ post}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year y (t CO₂e)

Calculation of methane flared or combusted (MD_y) can only be calculated for verification and shall be calculated as follows:

$$MD_y = BG_{burnt,y} \times w_{CH_4,y} \times D_{CH_4} \times FE \times GWP_{CH_4}$$

Equation 16

Where:

$BG_{burnt,y}$ = Biogas flared or combusted in year y (m³)

$w_{CH_4,y}$ = Methane content in biogas in the year y (volume fraction)

FE = Flare efficiency in the year y (fraction)

If flared or combusted biogas data is not available, alternatively, biogas flared or combusted (MD_y) also can be calculated as follows:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4} \times GWP_{CH_4}$$

Equation 17

Where:

EG_y = Total electricity generated from the recovered biogas in year y (MWh)

3600 = Conversion factor (1 MWh = 3600 MJ)

NCV_{CH_4} = NCV of methane (MJ/Nm³) (use default value: 35.9 MJ/Nm³)

EE_y = Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria:

- Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation.
- Default efficiency of 40 %.

Summary of emission reduction as per AMS-III.D is given in the table below:

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
29-September-2021 – 31-December-2021	9,944	662	1,802	7,480
2022	39,028	2,597	7,071	29,360
2023	39,028	2,597	7,071	29,360
2024	39,028	2,597	7,071	29,360
2025	39,028	2,597	7,071	29,360
2026	39,028	2,597	7,071	29,360
2027	39,028	2,597	7,071	29,360
01-January-2028 – 28-September-2028	29,084	1,935	5,269	21,880
Total	273,196	18,179	49,497	205,520
Average over the Crediting Period	39,028	2,597	7,071	29,360

Net GHG Emission Reduction and Removals as per AMS-I.D

In accordance with AMS-I.D., net emission reduction and removals has calculated by following formula:

$$ER_y = BE_y - PE_y - LE_y$$

Equation 18

ER_y = Emission reductions in year y (tCO₂)

BE_y = Baseline Emissions in year y (tCO₂)

PE_y = Project emissions in year y (tCO₂)

LE_y = Leakage emissions in year y (tCO₂)

As calculated in section 4.1 baseline emissions BE found as follows:

$$BE_y = EG_{pj,y} * EF_{grid,CM,y} = 11,700 \times 0.5552 = 6,495 \text{ tCO}_2/\text{year}$$

In accordance with paragraph 39 of AMS-I.D, project emission (PE_y) equals to zero and also leakage emission (LE_y) was assumed as 0.

As a result

$$ER_y = BE_y = 6,495 \text{ tCO}_2/\text{year}$$

Summary of emission reduction as per AMS-I.D is given in the table below:

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
29-September-2021 – 31-December-2021	1,672	0	0	1,672
2022	6,495	0	0	6,495
2023	6,495	0	0	6,495
2024	6,495	0	0	6,495
2025	6,495	0	0	6,495
2026	6,495	0	0	6,495
2027	6,495	0	0	6,495
01-January-2028 – 28-September-2028	4,823	0	0	4,823
Total	45,465	0	0	45,465
Average over the Crediting Period	6,495	0	0	6,495

Summary of net emission reductions as per AMS-III.D and AMS-I.D are given in the table below.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
29-September-2021 – 31-December-2021	11,598	662	1,802	9,134
2022	45,523	2,597	7,071	35,855
2023	45,523	2,597	7,071	35,855
2024	45,523	2,597	7,071	35,855
2025	45,523	2,597	7,071	35,855
2026	45,523	2,597	7,071	35,855
2027	45,523	2,597	7,071	35,855
01-January-2028 – 28 September 2028	33,923	1,936	5,269	26,718
Total	318,659	18,180	49,497	250,982
Average over the Crediting Period	45,523	2,597	7,071	35,855

6 MONITORING

6.1 Data and Parameters Available at Validation

Data / Parameter	GWP _{CH4}
Data unit	t CO ₂ e/t CH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC Fifth Assessment Report: Climate Change 2013 ¹³
Value applied:	28
Justification of choice of data or description of measurement methods and procedures applied	Default value IPCC Fifth Assessment Report (AR5)
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	D _{CH4}
Data unit	t CH ₄ / m ³ CH ₄
Description	Density of methane at normal conditions
Source of data	'Thermophysical properties of fluids. II. Methane, Ethane, Propane, Isobutane and Normal Butane' by B.A. Younglove, J.F. Ely Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Value applied:	0.00067

¹³ <https://www.ipcc.ch/report/ar5/syr/>

Justification of choice of data or description of measurement methods and procedures applied	Default values given in Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0
Purpose of Data	Calculation of baseline/project emissions
Comments	corrected to reference conditions

Data / Parameter	UF _b
Data unit	Dimensionless
Description	Model correction factor to account for model uncertainties (
Source of data	CDM Methodology AMS-III.D -Methane recovery in animal manure management systems
Value applied:	0.94
Justification of choice of data or description of measurement methods and procedures applied	Default values given in AMS-III.D.
Purpose of Data	Calculation of baseline/project emissions
Comments	-

Data / Parameter	MCF _j
Data unit	-
Description	Methane conversion factor for the baseline AWMS _j
Source of data	Obtained from 2019 IPCC Refinement, Table 10.17, p. 67 (uncovered anaerobic lagoon for 17.9°C, warm temperate moist)
Value applied:	0.73

Justification of choice of data or description of measurement methods and procedures applied	Default value. Uncovered anaerobic lagoon, Value for 17.9°C.
Purpose of Data	Used in project emission/baseline calculations
Comments	A conservativeness factor should be applied by multiplying MCF values (estimated as per the above bullet) with a value of 0.94 to account for the 20 per cent uncertainty in the MCF values as reported by IPCC

Data / Parameter	B _{0,LT}
Data unit	m ³ CH ₄ /kg-dm
Description	Maximum methane producing potential of the volatile solid generated for animal type LT
Source of data	Obtained from 2019 IPCC Refinement, Table 10.16,
Value applied:	0.24 for dairy cattle 0.39 for chicken layer
Justification of choice of data or description of measurement methods and procedures applied	Default values for maximum methane producing capacity
Purpose of Data	Used in project emission/baseline calculations
Comments	-

Data / Parameter	VS _{LT,y}
Data unit	kg -dm/animal/year
Description	The default value for the volatile solid excretion per day on a dry-matter basis for a defined livestock population

Source of data	Elaborated from 2019 IPCC Refinement, Table 10.13A p. 10.55 Middle East
Value applied:	1816,0575 dairy cattle 7.7745 for chicken layer
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of Data	Used in project emission/baseline calculations
Comments	-

Data / Parameter	$EF_{CH_4, default}$
Data unit	t CH ₄ leaked / t CH ₄ produced
Description	Default emission factor for the fraction of CH ₄ produced that leaks from the anaerobic digester
Source of data	2019 IPCC Refinement, Flesch et al. (2011) and Kurup (2003)
Value applied:	0.05
Justification of choice of data or description of measurement methods and procedures applied	As per Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0 UASB type digesters shall use the default of 0.05.
Purpose of Data	Calculation of project emissions
Comments	Applicable to the step “Determination of project emissions of methane from the anaerobic digester”

Data / Parameter	$f_{CH_4, default}$
Data unit	m ³ CH ₄ / m ³ biogas corrected to reference conditions

Description	Default value for the fraction of methane in the biogas
Source of data	The default value was taken from Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0 and the value was derived based on reported values from registered projects and research papers (Davidsson, 2007) ¹⁴
Value applied:	0.6
Justification of choice of data or description of measurement methods and procedures applied	As per Methodological Tool 14 Project and leakage emissions from anaerobic digesters Version 02.0, 0.6 is the default value.
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Republic of Türkiye Ministry of Energy in Emission Factor 2020
Value applied:	0.5552
Justification of choice of data or description of measurement methods and procedures applied	Calculated ex ante and fixed for the 7-year renewable crediting period.
Purpose of Data	Determination of baseline emissions
Comments	-

¹⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-14-v2.pdf>

Data / Parameter	EF _{CO2,transport}					
Data unit	kgCO ₂ /km					
Description	CO ₂ emission factor from fuel use due to transportation					
Source of data	Project and leakage emissions from road transportation of freight v.1.0					
Description of measurement methods and procedures applied	Distance between manure producer and the plant via Map applications					
Frequency of monitoring/recording	Annually					
Value applied:	<table><tr><th>Vehicle Class</th><th>Emission factor</th></tr><tr><td>Light Vehicle</td><td>6.37</td></tr></table>		Vehicle Class	Emission factor	Light Vehicle	6.37
Vehicle Class	Emission factor					
Light Vehicle	6.37					
Monitoring equipment	-					
QA/QC procedures applied	Default value					
Purpose of data	Project emission calculations					
Calculation method	For the calculation of emission factor, emission factor of “Project and leakage emissions from road transportation of freight” v1.0 is taken. The unit for emission factor is gCO₂/t km which means carrying emission of 1 ton of waste per kilometer. To find emission factor in accordance with the methodology this value is multiplied with 26 tone which makes highest possible emission factor to be conservative. Then the emission factor in unit of kgCO₂/km is calculated by following: 0.245 kgCO₂/t km * 26 t = 6.37 kgCO₂/km					
Comments	-					

Data / Parameter	TDL _{j,y}
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Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j or k in year y
Source of data	Methodological tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0)
Value applied:	3%
Justification of choice of data or description of measurement methods and procedures applied	According to the tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0), the project belongs to the case of Scenario A, use as default values of 3% for project emission since electricity consumption by all project electricity consumption sources is smaller than the electricity consumption of all baseline electricity consumption sources, and 3% for baseline emissions (Option 3 (b) on page 15 of AMS-III.D, version 21.0).
Purpose of Data	Calculation of project emissions from electricity consumption
Comments	-

Data / Parameter	NCV _{CH4}
Data unit	MJ/Nm ³
Description	Net calorific value of methane at reference conditions
Source of data	AMS-III.G. “Landfill methane recovery” (version 10.0).
Value applied	35.9
Justification of choice of data or description of measurement methods and procedures applied	Default Value
Purpose of Data	Calculation of emission reduction
Comments	-

Data / Parameter	EE _y
Data unit	%
Description	Energy conversion efficiency of the project equipment
Source of data	AMS-III.G. "Landfill methane recovery" (version 10.0).
Value applied	40%
Justification of choice of data or description of measurement methods and procedures applied	Default Value
Purpose of Data	Calculation of emission reduction
Comments	-

Data / Parameter	F _{ww,CH4,default}
Data unit	fraction
Description	Default factor representing the remaining CH ₄ production capacity of liquid digestate
Source of data	Tool14 "Project and leakage emissions from anaerobic digesters" (version 02.0).
Value applied	0.20
Justification of choice of data or description of measurement methods and procedures applied	Since the digesters are high-rate digesters, default value for high-rate digesters has selected.
Purpose of Data	Calculation of leakage emissions
Comments	-

Data / Parameter	W _{default}
Data unit	kg
Description	Default average animal weight of a defined population

Source of data	2019 IPCC Refinement , Volume 4, Chapter 10, Table 10A.5
Description of measurement methods and procedures applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	275 kg for dairy cattle 0.7 kg for chicken broiler
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	To calculate baseline emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	VS _{default}
Data unit	kg-VS/(1000 kg animal mass)/day
Description	Default value for the volatile solid excretion rate per day on a dry-matter basis for a defined livestock population
Source of data	2019 IPCC Refinement, Volume 4, Chapter 10, Table 10.13A
Description of measurement methods and procedures applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	10.7 kg-VS/(1000 kg animal mass)/day for dairy cattle 17.7 kg-VS/(1000 kg animal mass)/day for chicken broiler

Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	To calculate baseline emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	P _n
Data unit	Pa
Description	Total pressure at normal conditions
Source of data	Tool 08, version 03.0
Description of measurement methods and procedures applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	101,325
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	To calculate project emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	T _n
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Data unit	K
Description	Temperature at normal conditions
Source of data	Tool 08, version 03.0
Description of measurement methods and procedures applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	273.15
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	To calculate project emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	MM _i
Data unit	kg/kmol
Description	Molecular mass of greenhouse gas i
Source of data	Tool 08, version 03.0
Description of measurement methods and procedures applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	16.04

Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	To calculate project emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	R_u
Data unit	Pa.m ³ /kmol.K
Description	Universal ideal gases constant
Source of data	Tool 08, version 03.0
Description of measurement methods and procedures applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	0,008314
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	To calculate project emissions
Calculation method	N/A
Comments	N/A

6.2 Data and Parameters Monitored

Data / Parameter	EG _{PJ,y}															
Data unit	MWh/year															
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y															
Source of data	Electricity meters / EPIAS records															
Description of measurement methods and procedures applied	This parameter should be either monitored using bi-directional energy meter or calculated as difference between (a) the quantity of electricity supplied by the project plant/unit to the grid; and (b) the quantity of electricity the project plant/unit from the grid. In case it is calculated then the following parameters shall be measured: (a) The quantity of electricity supplied by the project plant/unit to the grid; and (b) The quantity of electricity delivered to the project plant/unit from the grid															
Frequency of monitoring/recording	Monthly															
Value applied:	Electricity generation estimated as 11,700 MWh ¹⁵ (<i>ex-ante value, will be monitored ex post</i>)															
Monitoring equipment	Electricity meters <table><tr><th>Equipment</th><th>Serial Number</th><th>Brand & Model</th><th>Accuracy Class</th><th>Date of Calibration</th></tr><tr><td>Main meter</td><td>80281239</td><td>Makel – C520 AMT.2556</td><td>C (0.5)</td><td>25/06/2021</td></tr><tr><td>Backup meter</td><td>80281229</td><td>Makel – C520 AMT.2556</td><td>C (0.5)</td><td>25/06/2021</td></tr></table>	Equipment	Serial Number	Brand & Model	Accuracy Class	Date of Calibration	Main meter	80281239	Makel – C520 AMT.2556	C (0.5)	25/06/2021	Backup meter	80281229	Makel – C520 AMT.2556	C (0.5)	25/06/2021
Equipment	Serial Number	Brand & Model	Accuracy Class	Date of Calibration												
Main meter	80281239	Makel – C520 AMT.2556	C (0.5)	25/06/2021												
Backup meter	80281229	Makel – C520 AMT.2556	C (0.5)	25/06/2021												
QA/QC procedures applied	Electricity meters are subject to regular (in accordance with stipulation of the meter supplier) maintenance and testing to ensure accuracy. The frequency of the calibration of the electricity meters is 10 years as per “Measuring Instruments Regulation”¹⁶															

¹⁵ Generation License

¹⁶ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

Purpose of data	This parameter is required for calculating baseline emissions from electricity generation due to waste treatment under the Project.
Calculation method	No calculation is needed measurement values are used.
Comments	-

Data / Parameter	EG _y
Data unit	MWh/year
Description	Quantity of gross electricity generated by the project plant
Source of data	SCADA
Description of measurement methods and procedures applied	This parameter is monitored using SCADA on the biogas genset
Frequency of monitoring/recording	Continuous measurement, hourly recording, monthly reported.
Value applied:	Will be monitored ex post
Monitoring equipment	PLC
QA/QC procedures applied	The device will be calibrated periodically by equipment manufacturer. The frequency of the calibration of the electricity meters is 10 years as per “Measuring Instruments Regulation”¹⁷
Purpose of data	This parameter is required for calculating emission reduction
Calculation method	No calculation is needed measurement values are used.
Comments	

¹⁷ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

Data / Parameter	$N_{da,y}$
Data unit	Number
Description	Number of days animal is alive in the farm in the year y
Source of data	Project proponents and farms
Description of measurement methods and procedures applied	Plant's log
Frequency of monitoring/recording	Annually, based on monthly records
Value applied:	365
Monitoring equipment	-
QA/QC procedures applied	-
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	Number of animal for type LT are kept same on average in accordance with facility's capacity and plan. In other words, mortality and growth rate is kept same. Therefore, no growth or deduction is expected in normal circumstances. Average number of animals are provided by statement of each facility.

Data / Parameter	$N_{p,y}$
Data unit	Number
Description	Number of animals produced annually of type LT for the year y
Source of data	Project proponents
Description of measurement methods and procedures applied	Plant's log
Frequency of monitoring/recording	Annually, based on monthly records
Value applied:	2500 diary cattle, 1,211,100 chicken layers (as per agreements signed with farmers)
Monitoring equipment	-
QA/QC procedures applied	Agreements signed with the farmers will be used for cross check
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	Number of animal for type LT are kept same on average in accordance with facility's capacity and plan. In other words, mortality and growth rate is kept same. Therefore, no growth or deduction is expected in normal circumstances. Average number of animals are provided by statement of each facility.

Data / Parameter	$MS\%_{BI,j}$
Data unit	%
Description	Fraction of manure handled in system j in the Project
Source of data	Project proponent

Description of measurement methods and procedures applied	Plant's log
Frequency of monitoring/recording	Annually
Value applied:	100% for Kayı Farm and Ali Aydın farm; for the rest, 50% of the produced manure treated by the proposed Project. <i>(as per agreements signed with farmers)</i>
Monitoring equipment	-
QA/QC procedures applied	Agreements signed with the farmers will be used for cross check
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	-

Data / Parameter	W_{site}
Data unit	kg
Description	Average animal weight of a defined livestock population at the project site
Source of data	Plant Logs
Description of measurement methods and procedures applied	Since IPCC values of VS are adjusted (2019 IPCC Refinement)
Frequency of monitoring/recording	-
Value applied:	465 kg for dairy cattle 1.5 kg for chicken layer
Monitoring equipment	-

QA/QC procedures applied	
Purpose of data	Baseline/emission reductions calculations
Calculation method	-
Comments	-

Data / Parameter	Q_{biogas}
Data unit	Nm ³ biogas
Description	Amount of biogas collected at the digester outlet in year y
Source of data	Flow meter
Description of measurement methods and procedures applied	The volumetric flow is measured by flow meters and data is shown in SCADA and recorded by SCADA operator daily.
Frequency of monitoring/recording	Continuously measurement by the flow meter. Data to be aggregated monthly and yearly
Value applied:	3,704,429 Nm ³ (<i>ex-ante value, will be monitored ex-post</i>)
Monitoring equipment	Flow meter (Please refer to Section 1.11 for the detailed information)
QA/QC procedures applied	The frequency of the calibration of the flow meter is 10 years as per "Measuring Instruments Regulation" ¹⁸
Purpose of data	Project emissions calculations
Calculation method	-
Comments	For the validation, the used formula is given below: $Q_{biogas} = \sum_{j,LT} (MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{BL,j}) / f_{CH4,default}$

¹⁸ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

	Actual value will be monitored ex post.
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Data / Parameter	W _{CH4}
Data unit	%
Description	Methane content in biogas in the year y
Source of data	Analyser
Description of measurement methods and procedures applied	The fraction of methane in the biogas should be measured with a continuous analyser
Frequency of monitoring/recording	Same frequency as the flow or, with periodical measurements at a 90/10 confidence/precision level by following the “Standard for sampling and surveys for CDM project activities and Programme of Activities”,
Value applied:	60% (<i>ex-ante value, will be monitored ex-post if required</i>)
Monitoring equipment	Analyser
QA/QC procedures applied	The frequency of the calibration of the gas analyzer is 10 years as per “Measuring Instruments Regulation” ¹⁹
Purpose of data	Emission reductions and project emissions calculations
Calculation method	-
Comments	-

Data / Parameter	FE
Data unit	%

¹⁹ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

Description	Flare efficiency
Source of data	-
Description of measurement methods and procedures applied	As per para.22 of the applied methodology, if the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. Flare unit will be used only emergency situations. Efficiency of flare unit is assumed to be 0% to be conservative. In other words, methane reduction from flare unit will not be claimed.
Frequency of monitoring/recording	-
Value applied:	0% for flare unit 100% for biogas engine
Monitoring equipment	-
QA/QC procedures applied	Regular maintenance will be carried out according with manufacturer recommendation to ensure optimal flare operation. It will be ensured that the temperature of the flare ($T_{EG,m}$) and the flow rate of the residual gas to the flare ($F_{RG,m}$) is within the manufacturer's specification for the flare ($SPEC_{flare}$).
Purpose of data	Project emissions calculations
Calculation method	-
Comments	-
Data / Parameter	$F_{CH4, RG, y}$
Data unit	t_{CH4}

Description	Mass of methane sent to flare in year y
Source of data	-
Description of measurement methods and procedures applied	The amount of biogas sent to flare is measured continuously by a flow meter.
Frequency of monitoring/recording	Continuous measurement
Value applied:	0 (to be monitored ex-post)
Monitoring equipment	-
QA/QC procedures applied	Regular maintenance will be carried out according with manufacturer recommendation. The frequency of the calibration of the flow meter is 10 years as per "Measuring Instruments Regulation" ²⁰
Purpose of data	Project emissions calculations
Calculation method	The amount of biogas sent to flare is measured continuously by a flow meter. Then, this amount is multiplied with fraction of methane in biogas, and density of methane to find mass flow of methane and then multiplied with duration of flare in use to find mass of methane sent to flare
Comments	-

Data / Parameter	Q_y and $Q_{res\ waste,y}$
Data unit	Tons
Description	Quantity manure brought to the plant
Source of data	On-site records
Description of measurement methods and procedures applied	On-site data sheets recorded monthly using weigh bridge.

²⁰ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

Frequency of monitoring/recording	Monthly
Value applied:	-
Monitoring equipment	Weight bridge
QA/QC procedures applied	Weighbridge will be subject to periodic calibration (in accordance with stipulation of the weighbridge supplier). The frequency of the calibration of the weighbridge is 10 years as per "Measuring Instruments Regulation"²¹
Purpose of data	Project emission calculations
Calculation method	-
Comments	For the validation, amount of waste per farm was calculated based on animal number and potential excretion rate by animal weight. In accordance with a study performed by the Ministry of Agriculture and Forestry ²² daily excretion rate is 4% of animal weight.

Data / Parameter	CT _y and CT _{res waste,y}
Data unit	Tons/truck
Description	Average truck capacity for transportation
Source of data	On-site measurement
Description of measurement methods and procedures applied	On-site data sheets recorded monthly using weigh bridge.
Frequency of monitoring/recording	Monthly

²¹ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

²²

[https://arastirma.tarimorman.gov.tr/tavukculuk/Belgeler/web%20English%20Doc/journal%20\(Dergimiz\)/Dergimiz%20Cilt%2011%20Sayi%201/Cilt%2011%20Sayi%201%20Makale%204%20Yeni%20Bir%20Teknoloji%20ile%20Kurutulan%20Tavuk%20D%C4%B1nC5%9Fk%C4%B1s%C4%B1n%C4%B1n.pdf](https://arastirma.tarimorman.gov.tr/tavukculuk/Belgeler/web%20English%20Doc/journal%20(Dergimiz)/Dergimiz%20Cilt%2011%20Sayi%201/Cilt%2011%20Sayi%201%20Makale%204%20Yeni%20Bir%20Teknoloji%20ile%20Kurutulan%20Tavuk%20D%C4%B1nC5%9Fk%C4%B1s%C4%B1n%C4%B1n.pdf)

Value applied:	26 tons/truck in average
Monitoring equipment	Weight bridge
QA/QC procedures applied	Weighbridge will be subject to periodic calibration (in accordance with stipulation of the weighbridge supplier). The frequency of the calibration of the weighbridge is 10 years as per "Measuring Instruments Regulation" ²³
Purpose of data	Project emission calculations
Calculation method	-
Comments	-

Data / Parameter	DAF _w and DAF _{res waste}
Data unit	km/truck
Description	Average incremental distance for raw solid or product transportation
Source of data	Map applicaitons
Description of measurement methods and procedures applied	Distance between manure producer and the plant via Map applications
Frequency of monitoring/recording	Annually
Value applied:	Please see section 4.2
Monitoring equipment	-
QA/QC procedures applied	-
Purpose of data	Project emission calculations

²³ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

Calculation method	-
Comments	-

Data / Parameter	nd _y
Data unit	Days
Description	Number of days that the animal manure management system was operational
Source of data	Project Proponent
Description of measurement methods and procedures applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	365
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	To calculate baseline emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	V _{t,db}
Data unit	m ³ dry gas/h
Description	Volumetric flow of the gaseous stream in time interval t on a dry basis

Source of data	Flow meter
Description of measurement methods and procedures applied	Flow meter results
Frequency of monitoring/recording	Continuous
Value applied:	Please refer to “Column D” in “Project Emissions” Excel spreadsheet
Monitoring equipment	Flow meter
QA/QC procedures applied	The frequency of the calibration of the flow meter is 10 years as per “Measuring Instruments Regulation” ²⁴
Purpose of data	To calculate project emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	$V_{i,t,db}$
Data unit	m ³ gas i/m ³ dry gas
Description	Volumetric fraction of greenhouse gas i in a time interval t on a dry basis
Source of data	Gas analyzer
Description of measurement methods and procedures applied	Gas analyzer results
Frequency of monitoring/recording	Continuous
Value applied:	Please refer to “Column E” in “Project Emissions” Excel spreadsheet

²⁴ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

Monitoring equipment	Gas analyzer
QA/QC procedures applied	The frequency of the calibration of the gas analyzer is 10 years as per “Measuring Instruments Regulation” ²⁵ .
Purpose of data	To calculate project emissions
Calculation method	N/A
Comments	N/A

6.3 Monitoring Plan

The monitoring methodology is based on directly measuring the biogas produced in the digesters, the biogas sent to the flare unit, and the electricity generated in the engines, as shown in the table below. The monitoring plan continuously measures the quantity and quality of biogas generated and flared. The main variables that need to be determined are the quantity of biogas burnt or gainfully used on a dry basis in year y ($BG_{burnt,y}$) and the quantity of methane used to generate electricity. The methodology also measures the energy generated using biogas ($EG_{PJ,y}$). However, if MDy is calculated through energy generation (by Equation 15), monitoring of w_{CH_4} , and $BG_{burnt,y}$ are not required.

From the monitoring methodology, it could be seen that there are the following main variables to be measured:

Table 2: Summary of Monitoring Plan

Parameter	Description
Q_{biogas}	Amount of biogas collected at the digester outlet in year y (Nm ³ biogas)
$F_{CH_4, RG, y}$	Amount of methane which is sent to the flare unit in year y
w_{CH_4}	Methane fraction in the biogas (%)
EG_y	Gross electricity generated (MWh)
$EG_{PJ, y}$	Net electricity delivered to the grid (MWh)

Responsibilities for data processing and management lie with Yılmazlar. Therefore, they established an emission reduction team. This team is responsible for monitoring all data required to estimate emission reductions. Life Enerji also assists VER Team concerning the monitoring

²⁵ <https://www.resmigazete.gov.tr/eskiler/2016/06/20160629-22.htm>

aspects of the Project. The plant manager has the main responsibility for collecting and archiving the data. According to the monitoring plan, qualified technicians monitor and record the data. All the technicians receive proper training to ensure that they understand their specific tasks and equipment handling. The records are double-checked by the General Manager of the Proposed Project, who is responsible for the accuracy and frequency of the measurements.

EPIAS stores the power generation data. The website of EPIAS is accessible by the Project Proponent with their unique user ID and password. Once accessed, the Project Proponents can retrieve electricity generation and consumption data. The Project Proponent uses the same data for invoicing purposes. The electricity generation data are reported on a monthly basis.

For the crosscheck, electricity distribution company stores reading data by its system named OSOS and accessible by the Project Proponent with their unique user ID and password.

Data Processing and collection

The monitoring equipment and other plant equipment are connected to a Programmable Logic Controller (PLC), which collects all the monitoring data and sends them to the SCADA system so that the operator can read the main parameters in real-time. A SCADA operator continuously read the data and record the read parameters. The main source for biogas and gross electricity generation data is the Project's PLC-SCADA system. The system automatically reads measurement devices and the data manually recorded. This data is used to calculate the total emission reductions.

1) Flow measurements

Gas flow is measured by flow meters, including the flow into the flare, generators and the total flow. These parameters are required to be normalised at 0°C and 1 atm, using actual gas temperature and pressure data to be measured with temperature and pressure transmitters. These in-built transmitters are integrated into each flow meter and equipped with a flow computer so that the values sent to the SCADA system have already been normalised (converted to normal cubic meters). Therefore, temperature and pressure data at each flowmeter do not need to be separately monitored. Methane content in the gas is measured by a gas analyser connected to the PLC-SCADA system.

Calibration: The flow sensors are calibrated according to the gas's specified temperature, pressure, and composition as per the manufacturer's recommendation. The equipment selected will allow dynamic compensation for these parameters, normalised to standard temperature, pressure, and gas composition. There will be a periodic verification according to the requirement of equipment specifications.

2) Gas Quality and efficiency of the flare

An enclosed flare unit determines methane destruction with an inbuilt flame detection unit. The project emissions from flaring and emission reductions from the destruction of biogas have been monitored at this point. The readings have been taken and recorded internally by compatible and

proper measurement devices with high data quality. The PLC system automatically converts the measured parameters to dry- basis and normal conditions. The instruments monitor parameters continuously. The emissions are calculated using the methodological tool "Project emissions from flaring" (v04.0).

The concentration of methane and oxygen in the gaseous stream and the exhaust gas of the flare are the parameters that are essential for the calculation of emission reductions and the safe and efficient operation of the system.

The concentration of methane and oxygen in the biogas stream is controlled by a common sample line installed in the main collection system piping and measured continuously by two separate analysers to measure methane, oxygen and carbon dioxide each. The sensors are designed to operate within specified temperature and pressure conditions.

The concentration of methane and oxygen in the exhaust gas stream is monitored by a common sample line installed in the upper section of the flare.

Calibration: Analysers are periodically calibrated according to the manufacturers and regulations on "Metering and Testing of Metering Systems"²⁶ of the Ministry of Industry and Technology recommendation. Calibration equipment will provide +/- 1% accuracy by volume.

3) Electricity Generation:

The electricity generation data is stored by EPIAS and electricity distribution company Osmangazi EDAŞ. The website of EPIAS and Osmangazi EDAŞ is accessible by the Project Proponent with their ID used for control and checks and invoicing purposes. The electricity generation data is reported on a monthly basis. The procedure involves the following tasks:

- Accessing the website of EPIAS
- Obtaining the electricity generation and consumption reports
- Store the data in electronic version on the computer of Yilmazlar on the plant
- Issue invoices to related distribution company based on the data published in EPIAS.

On the other hand, since the methane destruction can be calculation through electricity generation (by equation 11), using gross electricity generation is more accurate since it reflects destructed methane amount. EPIAS records include losses like transmission losses, so gross electricity generation read on SCADA of biogas genset is recorded and can be used for calculation of methane destructed.

A Backup Unit for Systematic Storage

i. Electricity Generation:

²⁶ See, http://www.sanayi.gov.tr/download/osgm/olcu_aletleri_muayene_yonetmelik.zip (page 2)

There are two measurement devices; a primary and a secondary (i.e. backup or check-meter) measurement device for quality assurance, to be used if the primary meter fails. Both the main and backup measurement devices measure and store the aggregate total electricity import and export for the whole Project in real-time. The electricity meters are under the responsibility of distribution company, and they are located in the sub-station, which also belongs to the distribution company.

The invoices are kept by the Project Proponent as hard copies. Furthermore, the EPIAS system stores the reports electronically, which is accessible to the Project Proponent whenever necessary. After the monthly reports are obtained from EPIAS, data is aggregated using a separate spreadsheet. The monthly generation and consumption data are entered into the spreadsheet, which calculates the emission reductions during the monitoring period.

ii. Methane Destruction:

Data Storage: The electronic monitoring system (PLC) will periodically read the data continuously and recorded manually by an operator. Written documents (e.g. equipment replacement protocols, accident logs, maintenance records, back-ups etc.) will be kept safely. Electronically backup of the data will be conducted on a daily basis. A hard copy backup of all relevant data will be printed out monthly. Calibration records for all instrumentation will be constantly collected and archived. All data and records required for verification are kept for two years after the project crediting period ends or the last issuance of VERs, whichever is later.

On the other hand, if electricity generation is used to calculate methane destruction, gross electricity generation on SCADA system can be used. The generations are recorded hourly, basis by operator manually. Moreover, cumulative electricity generation is stored on SCADA itself.

Calibration Procedures:

Maintenance and calibration of measurement devices: According to manufacturer specifications, all measurement devices has been purchased and maintained as specified in this report. All measurement devices that are used in monitoring is subject to a quality control procedure that include regular maintenance and calibration in agreement with legal and/or manufacturer requirements. According to the relevant regulation, "The Turkish Electricity Market Regulation Agency (EPDK) sets rules on the accuracy of electricity meters that are used by power plants feeding into the grid. The rules are part of the EPDK regulation 25056 from 22/03/2003²⁷. Also, according to Article 11 of this Communiqué, meters shall be in class of 0.5 for connected transformer less than 10 MVA, 0.5S for connected transformer 10 MVA and 100 MVA and 0.2S for connected transformer higher than 100 MVA and refers to compliance with International Electro technical Commission's norm EN 60687 . TEIAS, whose employees visit the plant for the meter readings monthly, is in charge of ensuring adherence to these rules. Calibration and maintenance procedures follow the requirements."

²⁷ <https://www.resmigazete.gov.tr/eskiler/2003/03/20030322.htm#5>

If any equipment malfunction or breakdown occurs, corrective actions are carried out in a timely manner to minimise the risk of emissions that are not intended. To ensure data quality, operational staff will be trained appropriately to take action effectively in such cases. The plant operator periodically inspects the plant to check visually if there are any apparent problems. In case of any findings, these are documented.

In case of an organisational change within the Project Plant, a qualified person will be assigned to carry out and manage the monitoring and verification procedures. In case of fire, earthquake or another similar emergency situation, the Project Proponent's data as backup stores will be used to maintain conservativeness with specific procedures. For extraordinary events where such backup data are unavailable, the amount of biogas consumed will be calculated using the power generation figures and engines' full-load electrical efficiencies by using conservative values and other assumptions to maintain conservativeness. Regarding the project emissions from flaring, data recording and aggregation and procedures for extraordinary events are the same as described above (i.e. the methane destruction component).

Monitoring will be carried out following the procedures set by the applicable methodology and all applicable tools of the above-explained parameters. The monitoring consists of:

Responsibilities for monitoring

The Project Proponent is responsible for the installed equipment operation and maintenance. The Project Proponent is also responsible for the administration of the data, setting up a VER team who will be responsible for monitoring all data required to estimate emission reductions.

Registration of the monitored data

For Yilmazlar BPP Project, VER-Team will be responsible for monitoring equipment for quality assurance. The data measured by the monitoring equipment will be stored and will be processed into a monitoring report, which the Project Proponent will submit. All the monitored data will be stored during the crediting period and for at least two years after the end of the crediting period, whichever occurs later.

Corrective actions and emergency preparedness

The VER-Team will regularly check the monitoring system for errors. In the case of errors, corrective actions will be undertaken by the VER-Team, or if required, by the monitoring equipment supplier.

QA/QC procedure

Strong quality assurance and quality control procedures will be taken to monitor the equipment and data collection. Equipment and facilities will be subject to a regular maintenance and testing regime to ensure accuracy following the supplier's manual. In case of data deficiency or correct data reading, the data will be cross-checked with other parameters and data leading to the most conservative emission reductions will be considered for calculating the emission reductions.

Measuring of exported and imported electricity

The main meter for monitoring the electricity import and export is at the grid connection. Monthly protocols based on the readings confirmed by TEİAŞ and the project participant are prepared by the end of each month and stored. These protocols are also the basis of invoicing for the electricity sales from the proposed Project. The electricity meters are bi-directional and will measure both the imported and exported electricity from and to the grid.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Data / Parameter	EGPJ,y																									
Data unit	MWh/year																									
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y																									
Value applied:	Total Amount is 8,992 MWh <table><tr><th>Vintage</th><th>Value (MWh)</th></tr><tr><td>29 September 2021 – 31 December 2021</td><td>534.43</td></tr><tr><td>01 January 2022 – 31 December 2022</td><td>6,459.28</td></tr><tr><td>01 January 2023 – 31 March 2023</td><td>1,997.95</td></tr><tr><td>Total</td><td>8,992</td></tr></table> Monitoring Equipment (electricity meter) specifications <table><tr><th>Equipment</th><th>Serial Number</th><th>Brand & Model</th><th>Accuracy Class</th><th>Date of Calibration</th></tr><tr><td>Main meter</td><td>80281239</td><td>Makel – C520 AMT.2556</td><td>C (0.5)</td><td>25/06/2021</td></tr><tr><td>Backup meter</td><td>80281229</td><td>Makel – C520 AMT.2556</td><td>C (0.5)</td><td>25/06/2021</td></tr></table>	Vintage	Value (MWh)	29 September 2021 – 31 December 2021	534.43	01 January 2022 – 31 December 2022	6,459.28	01 January 2023 – 31 March 2023	1,997.95	Total	8,992	Equipment	Serial Number	Brand & Model	Accuracy Class	Date of Calibration	Main meter	80281239	Makel – C520 AMT.2556	C (0.5)	25/06/2021	Backup meter	80281229	Makel – C520 AMT.2556	C (0.5)	25/06/2021
Vintage	Value (MWh)																									
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Total	8,992																									
Equipment	Serial Number	Brand & Model	Accuracy Class	Date of Calibration																						
Main meter	80281239	Makel – C520 AMT.2556	C (0.5)	25/06/2021																						
Backup meter	80281229	Makel – C520 AMT.2556	C (0.5)	25/06/2021																						
Comments	The data is monitored by the bi-directional electricity meter that installed at the outlet of the onsite substation. Meter readings are read and recorded by Osmangazi EDAŞ (Distribution Company) and EPIAS. Since the electricity transferred via energy transmission line, EPIAS deduct this amount and EPIAS system shows invoice-based reading by excluding transmission losses EPIAS data will be the main data and Osmangazi EDAŞ readings will be used for cross-check.																									

Data / Parameter	EG _y		
Data unit	MWh/year		
Description	Quantity of gross electricity generation		
Value applied:	Total amount is 9,123 MWh		
	Vintage		Value (MWh)
	29 September 2021 – 31 December 2021		592,43
	01 January 2022 – 31 December 2022		6,531.56
	01 January 2023 – 31 March 2023		1,999.47
	Total		9,123
	Monitoring Equipment (electricity meter) specifications		
Equipment	Brand & Model	Voltage Accuracy Class	Power Accuracy Class
PLC	Easygen 3500XT-P1	Class 0.5	Class 1.0
Comments	The data is monitored by the PLC that installed at the alternator of the biogas genset.		
	For methane reduction calculations, gross generation is used since energy generation reasoned by methane combustion is claimed and amount of transmission or any other losses should be excluded.		

Data / Parameter	N _{da,y}
Data unit	days
Description	Number of days animal is alive in the farm in the year y
Value applied:	365
Comments	Number of days animal is alive is higher than 365 days but, number of animals are kept same by balancing death and

	growing rates in accordance with farm capacity which is also conservative.
--	--

Data / Parameter	$N_{p,y}$
Data unit	Number
Description	Number of animals produced annually of type LT for the year y
Value applied:	2500 diary cattle, 1,211,100 chicken layers
Comments	-

Data / Parameter	W_{site}
Data unit	kg
Description	Average animal weight of a defined livestock population at the project site
Value applied:	465 for diary cattle, 1.5 for chicken layer
Comments	For chicken layer, there are 11 farm owners stated wight of animal. To be conservative lowest weight is used for all.

Data / Parameter	$MS\%_{BI,j}$
Data unit	%
Description	Fraction of manure handled in system j in the Project
Value applied:	100% for Kayı Farm and Ali Aydın farm, For the rest, 50% of the produced manure treated by the proposed Project.
Monitoring equipment	-

Comments	-
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Data / Parameter	Q_{biogas}										
Data unit	Nm ³ biogas										
Description	Amount of biogas collected at the digester outlet in year y										
Value applied:	Total biogas amount is 5,085,085 Nm³ <table border="1"> <thead> <tr> <th>Vintage</th><th>Biogas Volume (Nm³)</th></tr> </thead> <tbody> <tr> <td>29 September 2021 – 31 December 2021</td><td>298,417.71</td></tr> <tr> <td>01 January 2022 – 31 December 2022</td><td>3,601,147.36</td></tr> <tr> <td>01 January 2023 – 31 March 2023</td><td>1,185,520.03</td></tr> <tr> <td>Total</td><td>5,085,085</td></tr> </tbody> </table>	Vintage	Biogas Volume (Nm ³)	29 September 2021 – 31 December 2021	298,417.71	01 January 2022 – 31 December 2022	3,601,147.36	01 January 2023 – 31 March 2023	1,185,520.03	Total	5,085,085
Vintage	Biogas Volume (Nm ³)										
29 September 2021 – 31 December 2021	298,417.71										
01 January 2022 – 31 December 2022	3,601,147.36										
01 January 2023 – 31 March 2023	1,185,520.03										
Total	5,085,085										
Comments	Data is used to calculate project emissions caused from physical leakage										

Data / Parameter	FE
Data unit	%
Description	Flare efficiency
Value applied:	0% for flare unit 100% for biogas engine
Comments	-

Data / Parameter	$F_{\text{CH}_4, \text{RG}, y}$
Data unit	tCH ₄
Description	Mass of methane sent flare in year y

Value applied:	Total methane flared in flare unit is: 20 tons.	
	Vintage	Methane amount (ton)
	29 September 2021 – 31 December 2021	0
	01 January 2022 – 31 December 2022	14
	01 January 2023 – 31 March 2023	6
	Total	20
Comments	-	

Data / Parameter	Q_y/CT_y and $Q_{res\ waste,y}/CT_{res\ waste,y}$		
Data unit	laps		
Description	Number of laps that trucks performed in year y		
Value applied:	Vintage	Farm Group 1 (Farms No 1 to 10)	Farm Group 2 (Farm No 11)
	29 September 2021 – 31 December 2021	11	3
	01 January 2022 – 31 December 2022	426	73
	01 January 2023 – 31 March 2023	148	30
Comments	Actually Q_y & $Q_{res\ waste,y}$ and CT_y & $CT_{res\ waste,y}$ should have been monitored as per the Tool and this amount is used to calculate number of laps a truck performed in year y. However, plant logs records number of laps directly which more accurate. On the other hand, when one log is available for farm no's 1 to 10, another log is recorded for Farm 11.		

Data / Parameter	DAF _w and DAF _{res waste}
Data unit	km/truck
Description	Average incremental distance for raw solid or product transportation
Value applied:	For DAF_w, 50 km is taken for farm no's 1 to 10, and 200 km is taken for Farm 11. DAF_{res waste} is 0 since the liquid waste is pumped to cropland nearby and very low amount of solid waste is produced and this is taken by villagers near the plant so it is negligible.
Comments	When one log is available for farm no's 1 to 10, another log is recorded for Farm 11. Therefore, longest distance is assumed for 1 st group of farms (Farm 1-10) to be conservative.

Data / Parameter	nd _y
Data unit	Days
Description	Number of days that the animal manure management system was operational
Source of data	Project Proponent
Description of measurement methods and procedures applied	N/A
Frequency of monitoring/recording	N/A
Value applied:	365
Monitoring equipment	N/A
QA/QC procedures applied	N/A
Purpose of data	To calculate baseline emissions

Calculation method	N/A
Comments	N/A

Data / Parameter	$V_{t,db}$
Data unit	m ³ dry gas/h
Description	Volumetric flow of the gaseous stream in time interval t on a dry basis
Source of data	Flow meter
Description of measurement methods and procedures applied	Flow meter results
Frequency of monitoring/recording	Continuous
Value applied:	Please refer to “Column D” in “Project Emissions” Excel spreadsheet
Monitoring equipment	Flow meter
QA/QC procedures applied	Calibration of the flow meter
Purpose of data	To calculate project emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	$V_{i,t,db}$
Data unit	m ³ gas i/m ³ dry gas
Description	Volumetric fraction of greenhouse gas i in a time interval t on a dry basis

Source of data	Gas analyzer
Description of measurement methods and procedures applied	Gas analyzer results
Frequency of monitoring/recording	Continuous
Value applied:	Please refer to “Column E” in “Project Emissions” Excel spreadsheet
Monitoring equipment	Gas analyzer
QA/QC procedures applied	Calibration of the gas analyzer
Purpose of data	To calculate project emissions
Calculation method	N/A
Comments	N/A

7.2 Baseline Emissions

Baseline emission reductions are calculated in accordance with AMS-III.D (Version 21.0) and AMS-I.D. (Version 18.0). The baseline emissions are calculated by the following equation:

$$BE_y = BE_{CH_4,y} + BE_{electricity,y} \quad \text{Equation 19}$$

Where;

BE_y = Baseline emissions in year y (tCO₂e/y)

$BE_{CH_4,y}$ = Baseline emissions of methane from the methane in year y (tCO₂e/y)

$BE_{electricity,y}$ = Baseline emissions associated with electricity displacement y (tCO₂e/y)

Determination of Baseline Emissions from AMS-III.D. ($BE_{CH_4,y}$)

As per AMS-III.D. (version 21) baseline emissions calculated by following formula.

$$BE_y = GWP_{CH_4} \times D_{CH_4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{BL,j} \quad \text{Equation 20}$$

Where;

BE_y	=	Baseline emissions in year y (t CO ₂ e)
GWP_{CH_4}	=	Global Warming Potential (GWP) of CH ₄ applicable to the crediting period (t CO ₂ e/t CH ₄)
D_{CH_4}	=	CH ₄ density (0.00067 t/m ³ at room temperature (20 °C) and 1 atm pressure)
LT	=	Index for all types of livestock
j	=	Index for animal manure management system
MCF_j	=	Annual methane conversion factor (MCF) for the baseline animal manure management system j
$B_{0,LT}$	=	Maximum methane producing potential of the volatile solid generated for animal type LT (m ³ CH ₄ /kg-dm)
$N_{LT,y}$	=	Annual average number of animals of type LT in year y (numbers)
$VS_{LT,y}$	=	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)
$MS\%_{BI,j}$	=	Fraction of manure handled in baseline animal manure management system j
UF_b	=	Model correction factor to account for model uncertainties (0.94)

Details of the calculations are given in section 4.1 of the report and the emission reduction excel sheet. As a result of the calculations, achieved baseline emissions are given in the table below:

Months	Baseline Emissions (BE _{y,ex post})
Sep-21	3,252
Oct-21	3,252
Nov-21	3,252
Dec-21	3,252
Sum 2021	13,008
Jan-22	3,252
Feb-22	3,252
Mar-22	3,252
Apr-22	3,252
May-22	3,252
Jun-22	3,252
Jul-22	3,252
Aug-22	3,252

Sep-22	3,252
Oct-22	3,252
Nov-22	3,252
Dec-22	3,252
Sum 2022	39,024
Jan-23	3,252
Feb-23	3,252
Mar-23	3,252
Sum 2023	9,756
Total Sum 29/09/2021- 31/03/2023	61,788

As per AMS-III.D, the actual emission reduction achieved for Type III component of the project (which means not including emission reduction from Type I component, i.e. methane power generation) during the crediting period will be calculated using the amount of methane recovered and destroyed/gainfully used by the Project, calculated as:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH_4} \times EE_y} \times D_{CH_4} \times GWP_{CH_4}$$

Equation 21

Where:

EG_y = Gross Electricity generation in year y (MWh)

3600 = Conversion factor (1 MWh = 3600 MJ)

NCV_{CH_4} = NCV of methane (MJ/Nm³) use default value: 35.9 MJ/Nm³

EE_y = Energy Conversion Efficiency of the project equipment determined from one of the following options:

- Specification provided by the equipment manufacturer specifically for biogas fuel only if the equipment is designed to utilize biogas as fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation
- Default efficiency of 40 per cent

Details of the calculations are given in the emission reduction excel sheet. As a result of the calculations, summary of methane captured and gainfully destroyed (MD_y) are given in the table below:

Months	Net Electricity Generations (EG _y , MWh)	Methane captured and gainfully destroyed (MD _y)
<i>Sep-21</i>	19.91	93
<i>Oct-21</i>	64.27	302
<i>Nov-21</i>	145.68	685
<i>Dec-21</i>	304.58	1,432
Sum 2021	534.44	2,512
<i>Jan-22</i>	310.98	1,462
<i>Feb-22</i>	315.12	1,482
<i>Mar-22</i>	415.84	1,955
<i>Apr-22</i>	559.79	2,632
<i>May-22</i>	604.62	2,843
<i>Jun-22</i>	545.17	2,563
<i>Jul-22</i>	569.67	2,679
<i>Aug-22</i>	548.75	2,580
<i>Sep-22</i>	606.54	2,852
<i>Oct-22</i>	712.53	3,351
<i>Nov-22</i>	619.56	2,913
<i>Dec-22</i>	648.86	3,051
Sum 2022	6,457.43	30,363
<i>Jan-23</i>	653.30	3,072
<i>Feb-23</i>	644.35	3,030
<i>Mar-23</i>	700.35	3,293
Sum 2023	1,998.00	9,395
Total Sum 29/09/2021- 31/03/2023	8,990	42,270

7.3 Project Emissions

Project Emissions as per AMS-III.D.

Project emissions of the Project as per AMS.III.D (version 21.0) is calculated by the following formula:

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

Equation 22

Where:

PE_y	=	Project emissions in year y (t CO ₂ e)
$PE_{PL,y}$	=	Emissions due to physical leakage of biogas in year y (t CO ₂ e)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the biogas stream in the year y (t CO ₂ e)
$PE_{power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO ₂ e)
$PE_{transport,y}$	=	Emissions from incremental transportation in the year y (t CO ₂ e)
$PE_{storage,y}$	=	Emissions from the storage of manure (t CO ₂ e)

Project Emissions from Physical Leakage ($PE_{PL,y}$)

In accordance with AMS-III.D. paragraph 21 (b), physical leakage of anaerobic digesters are calculated by following formula as per Tool 14 “Project and leakage emissions from anaerobic digesters”.

$$PE_{PL,y} = PE_{CH_4,y} = Q_{CH_4,y} \times EF_{CH_4,default} \times GWP_{CH_4}$$

Equation 23

Where:

$PE_{CH_4,y}$	=	Project emissions associated with the anaerobic digester in year y (t CO ₂ e)
$Q_{CH_4,y}$	=	Quantity of methane produced in the anaerobic digester in year y (t CH ₄)
$EF_{CH_4,default}$	=	Project emissions from fossil fuel consumption associated with the anaerobic digester in year y (t CO ₂ e)
GWP_{CH_4}	=	Project emissions of methane from the anaerobic digester in year y (t CO ₂ e)

And $Q_{CH_4,y}$ is calculated as follows:

$$Q_{CH_4,y} = Q_{biogas,y} \times f_{CH_4,default} \times P_{CH_4}$$

Equation 24

$Q_{CH_4,y}$	=	Quantity of methane produced in the digester in year y (t CH ₄)
$Q_{biogas,y}$	=	Amount of biogas collected at the digester outlet in year y (Nm ³ biogas)
$f_{CH_4,default}$	=	Default value for the fraction of methane in the biogas (m ³ CH ₄ / m ³ biogas)

$$\rho_{CH_4} = \text{Density of methane at normal conditions (t CH}_4\text{ / Nm}^3\text{ CH}_4\text{)}$$

Details of calculations of the project emissions from physical leakage is given in emission reduction excel sheet. Summary of $PE_{PL,y}$ is given in the table below:

Months	Biogas Volume (Q_{biogas} Nm ³)	Fraction of methane ($f_{CH_4, default}$, %)	Density of Methane (D_{CH_4} , t/m ³)	Methane Emission Factor ($EF_{CH_4, default}$, tCO ₂)	Global Warming Potential of Methane (GWP_{CH_4})	Project Emissions from physical Leakage (PE_{PL} , tCO ₂)
Sep-21	3,075.19	0.6	0.00067	0.05	28	2
Oct-21	39,649.80	0.6	0.00067	0.05	28	23
Nov-21	85,341.00	0.6	0.00067	0.05	28	49
Dec-21	170,351.72	0.6	0.00067	0.05	28	96
Sum 2021	298,417.71	0.6	0.00067	0.05	28	170
Jan-22	177,312.58	0.6	0.00067	0.05	28	100
Feb-22	177,232.45	0.6	0.00067	0.05	28	100
Mar-22	218,215.85	0.6	0.00067	0.05	28	123
Apr-22	311,287.78	0.6	0.00067	0.05	28	176
May-22	339,210.14	0.6	0.00067	0.05	28	191
Jun-22	300,604.20	0.6	0.00067	0.05	28	170
Jul-22	313,616.55	0.6	0.00067	0.05	28	177
Aug-22	307,812.82	0.6	0.00067	0.05	28	174
Sep-22	319,613.41	0.6	0.00067	0.05	28	180
Oct-22	369,184.54	0.6	0.00067	0.05	28	208
Nov-22	391,497.04	0.6	0.00067	0.05	28	221
Dec-22	375,560.00	0.6	0.00067	0.05	28	212
Sum 2022	3,601,147.36	0.6	0.00067	0.05	28	2032
Jan-23	381,269.43	0.6	0.00067	0.05	28	215
Feb-23	383,143.65	0.6	0.00067	0.05	28	216
Mar-23	421,106.95	0.6	0.00067	0.05	28	237
Sum 2023	1,185,520.03	0.6	0.00067	0.05	28	668
Total Sum 29/09/2021- 31/03/2023	5,085,085	0.6	0.00067	0.05	28	2,870

Project Emissions from Flaring ($PE_{flare,y}$)

As per AMS-III.D. version 21, paragraph 22, "If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%." Since the methane is used to generate electricity, efficiency is taken is 100% for the biogas combusted in biogas genset. On the other hand, there is a flare unit that is used in emergency situations. Flare efficiency for flare unit is taken as 0%. Project

emissions from flaring is calculated as per Tool 06 “Project emissions from flaring” version 4.0. For the calculation following formula is used:

$$PE_{flare,y} = GWP_{CH_4} \times \sum_{m=1}^{525600} F_{CH_4, RG, m} \times (1 - \eta_{flare, m}) \times 10^{-3}$$

Equation 25

Where:

$PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO₂e)

$F_{CH_4, RG, m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare, m}$ = Flare efficiency in minute m

In according to calculations, biogas sent to flare and project emissions from flaring is given in the table below.

Months	Q _{biogas, sent to flare, y} (m ³)	Fraction of methane (f _{CH₄, default, %})	Density of Methane (D _{CH₄} , t/m ³)	Mass flow of methane (F _{CH₄, RG, y} ton)	GWP _{CH₄}	Flare Efficiency (η _{flare} , %)	Project Emission from flare (PE _{flare, y} tCO ₂ e)
Sep-21	0	0.6	0.00067	0	28	0	0
Oct-21	0	0.6	0.00067	0	28	0	0
Nov-21	0	0.6	0.00067	0	28	0	0
Dec-21	0	0.6	0.00067	0	28	0	0
Sum/Avg 2021	0	0.6	0.00067	0	28	0	0
Jan-22	0	0.6	0.00067	0	28	0	0
Feb-22	0	0.6	0.00067	0	28	0	0
Mar-22	0	0.6	0.00067	0	28	0	0
Apr-22	0	0.6	0.00067	0	28	0	0
May-22	2500	0.6	0.00067	1	28	0	29
Jun-22	1404	0.6	0.00067	1	28	0	16
Jul-22	2370	0.6	0.00067	1	28	0	27
Aug-22	12304	0.6	0.00067	5	28	0	139
Sep-22	2431	0.6	0.00067	1	28	0	28
Oct-22	0	0.6	0.00067	0	28	0	0
Nov-22	9246	0.6	0.00067	4	28	0	105
Dec-22	3495	0.6	0.00067	1	28	0	40
Sum/Avg 2022	33750	0.6	0.00067	14	28	0	384
Jan-23	700	0.6	0.00067	0	28	0	8
Feb-23	5720	0.6	0.00067	2	28	0	65
Mar-23	9286	0.6	0.00067	4	28	0	105
Sum/Avg 2023	15706	0.6	0.00067	6	28	0	178

Total Sum 29/09/2021- 31/03/2023	49,456	0.6	0.00067	20	28	0.000	562
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For the calculation, flow rate of biogas sent to flare is monitored continuously by a flow meter, and then it is multiplied with default fraction of methane, density of methane and global warming potential of methane to calculate project emission from flaring.

Project Emissions from Fossil Fuel Consumption and Electricity Use ($PE_{Power,y}$)

Since the Project involves electricity generation, it is expected that the plant will generate more electricity than it consumes. Moreover, net electricity generation and emission reduction due to renewable energy generation will be claimed under AMS-I.D. v18.0 and energy consumption will be subtracted from energy generated and EPIAS data has already shows transmission and distribution losses for the electricity generated to grid and consumed from the grid. Therefore, PE_{power} is 0.

Project Emissions from Incremental Transportation ($PE_{transp,y}$)

As per AMS-III.D. version 21, project emissions from incremental transportation is calculated by formula as per AMS-III.AO version 1.0.

$$PE_{transp,y} = (Q_y / CT_y) * DAF_w * EF_{CO2,transport} + (Q_{res-waste,y} / CT_{res-waste,y}) * DAF_{res-waste} * EF_{CO2,transport} \quad \text{Equation 26}$$

Where:

- Q_y = Quantity of raw waste/manure treated and/or wastewater co-digested in the year y (tonnes)
- CT_y = Average truck capacity for transportation (tonnes/truck)
- DAF_w = Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck)
- $EF_{CO2,transport}$ = CO_2 emission factor from fuel use due to transportation (kg CO_2 /km, IPCC default values or local values may be used)
- $Q_{res-waste,y}$ = Quantity of residual waste produced in year y (tonnes)
- $CT_{res-waste,y}$ = Average truck capacity for residual waste transportation (tonnes/truck)
- $DAF_{res-waste}$ = Average distance for residual waste transportation (km/truck)

Instead of monitoring of Q_y , and CT_y , project proponent has records to each lap of transportation data which stands for Q_y/CT_y . On the other hand, large proportion of residual waste is generated as liquid phase and it is directly pumped to croplands nearby. Small proportion of was is generated in solid waste and it taken by villagers to be used as fertilizer. Since the distance of village is less than 0.5km and amount of waste low, project emission from incremental transportation for residual waste is neglected.

Farms are grouped, and lap per truck is recorded as per farm groups. Farms and distances were given in table below:

Group No	Name of the Farm	Distance to the plant (km)	Round distance (km)
Group 0	Kayı Farm	0	0
Group 1	As Tarım	18	36
Group 1	Baş Tarım	25	50
Group 1	Özkantar Gıda	25	50
Group 1	Halil Yeşil Yumurta Üretim	25	50
Group 1	Yoncalı Gıda	23	46
Group 1	Yumurtalar Tarım	22	44
Group 1	Emre Kayner	24	48
Group 1	Halil Küçük	24	48
Group 1	Saydam Tarım	24	48
Group 1	Bayraktar	24	48
Group 2	Ali Aydın	100	200

Since the records of trucks are taken in accordance with the group no, longest distance among group no is taken as incremental distance to be conservative. Therefore, incremental distance per farm group is given in the table below:

Group No	Distance to the plant (km)	Round distance (km)
Group No 0	0	0
Group No 1	25	50
Group No 2	100	200

As a result details of calculations of the project emissions from incremental transportation is given in emission reduction excel sheet. Summary of $PE_{transp,y}$ is given in table below.

Months	Number of Laps a truck performed for Farm Group 1 ($Q_{y,1} / CT_{y,1}$)	Number of Laps a truck performed for Farm Group 2 ($Q_{y,2} / CT_{y,2}$)	Distance to Farm Group 1 (DAF _{w,1}) (km)	Distance to Farm Group 2 (DAF _{w,2}) (km)	CO ₂ Emission Factor due to transportation (EF _{CO₂,transport}) (kgCO ₂ /km)	Number of Laps truck performed for Farm Group 2 ($Q_{res,waste} / CT_{res waste}$)	Distance to Farm Group 1 (DAF _{res,waste}) (km)	Project Emissions from Incremental Transportation ($PE_{transp,y}$) (tCO ₂)
Sep-21	0	0	50	200	6.37	0	0	0
Oct-21	0	0	50	200	6.37	0	0	0
Nov-21	0	0	50	200	6.37	0	0	0
Dec-21	11	3	50	200	6.37	0	0	8

Sum/Avg 2021	11	3	50	200	6.37	0	0	8
Jan-22	0	4	50	200	6.37	0	0	6
Feb-22	0	5	50	200	6.37	0	0	7
Mar-22	3	0	50	200	6.37	0	0	1
Apr-22	25	0	50	200	6.37	0	0	8
May-22	25	2	50	200	6.37	0	0	11
Jun-22	50	9	50	200	6.37	0	0	28
Jul-22	47	7	50	200	6.37	0	0	24
Aug-22	61	8	50	200	6.37	0	0	30
Sep-22	57	9	50	200	6.37	0	0	30
Oct-22	59	10	50	200	6.37	0	0	32
Nov-22	44	9	50	200	6.37	0	0	26
Dec-22	55	10	50	200	6.37	0	0	31
Sum/Avg 2022	426	73	50	200	6.37	0	0	234
Jan-23	58	10	50	200	6.37	0	0	32
Feb-23	73	8	50	200	6.37	0	0	34
Mar-23	17	12	50	200	6.37	0	0	21
Sum/Avg 2023	148	30	50	200	6.37	0	0	87
Total Sum/Avg. 29/09/2021-31/03/2023	585	106	50	200	6.37	0	0	329

Project Emissions from Storage of Manure ($PE_{storage,y}$)

Incoming waste is utilized within 24 hours. Therefore, project emissions from storage of manure is 0.

Summary of Project Emissions ($PE_{ex\ post,y}$)

Summary of ex-post project emissions are given in the table below:

Months	Project Emissions from Physical Leakage ($PE_{PL,y}$)	Project Emissions from Flaring ($PE_{flare,y}$)	Project Emissions from Fossil Fuel Usage ($PE_{Power,y}$)	Project Emissions from Incremental Transportation ($PE_{transp,y}$)	Project Emissions from Storage ($PE_{Storage,y}$)	Sum of Project Emissions ($PE_{ex-post,y}$)
Sep-21	2	0	0	0	0	2
Oct-21	23	0	0	0	0	23
Nov-21	49	0	0	0	0	49
Dec-21	96	0	0	8	0	104
Sum 2021	170	0	0	8	0	178
Jan-22	100	0	0	6	0	106
Feb-22	100	0	0	7	0	107
Mar-22	123	0	0	1	0	124
Apr-22	176	0	0	8	0	184

May-22	191	29	0	11	0	231
Jun-22	170	16	0	28	0	214
Jul-22	177	27	0	24	0	228
Aug-22	174	139	0	30	0	343
Sep-22	180	28	0	30	0	238
Oct-22	208	0	0	32	0	240
Nov-22	221	105	0	26	0	352
Dec-22	212	40	0	31	0	283
Sum 2022	2032	384	0	234	0	2,650
Jan-23	215	8	0	32	0	255
Feb-23	216	65	0	34	0	315
Mar-23	237	105	0	21	0	363
Sum 2023	668	178	0	87	0	933
Total Sum 29/09/2021- 31/03/2023	2,870	562	0	329	0	3,761

7.4 Leakage

In accordance with AMS-III.D. the leakage emission of the project is calculated following the relevant procedure in the methodological tool “Project and leakage emissions from anaerobic digesters”, version 02.0 and formula in the tool is given below:

$$LE_{AD,y} = LE_{storage,y} + LE_{comp,y}$$

Equation 27

Where:

- $LE_{AD,y}$ = Project emissions from electricity consumption in year y (t CO₂ / yr)
- $LE_{storage,y}$ = Leakage emissions associated with storage of digestate in year y (t CO₂e)
- $LE_{comp,y}$ = Leakage emissions associated with composting digestate in year y (t CO₂e)

In the Project, there is no composting and solid digestate amount is low and not stored. However, liquid digestate is stored for a while. Therefore, there is a leakage emission resulted from storage of liquid digestate. Leakage emission resulted from storage of liquid digestate is calculated by the following formula.

$$LE_{storage,y} = F_{ww,CH_4,default} \times Q_{CH_4,y} \times GWP_{CH_4}$$

Equation 28

Where:

- $LE_{storage,y}$ = Leakage emissions associated with storage of digestate in year y (t CO₂e)

$F_{\text{ww,CH}_4,\text{default}}$ = Default factor representing the remaining methane production capacity of liquid digestate (fraction)

$Q_{\text{CH}_4,y}$ = Quantity of methane produced in the digester in year y (t CH₄)

GWP_{CH_4} Global warming potential of CH₄ (t CO₂ / t CH₄)

Since the anaerobic digesters in the project are high rate digesters, Default factor representing the remaining methane production capacity of liquid digestate ($F_{\text{ww,CH}_4,\text{default}}$) equals to 0.15 as per Tool 14 “Project and leakage emissions from anaerobic digesters” version 02.0. As a result of this, leakage emissions are given in the table below.

Months	Biogas Volume ($Q_{\text{biogas},}$) (Nm ³)	Fraction of methane ($f_{\text{CH}_4,\text{default},}$) (%)	Density of Methane ($D_{\text{CH}_4,}$) (t/m ³)	Quantity of Methane Volume ($Q_{\text{CH}_4,}$) (tCH ₄)	Remaining Methane Production Capacity ($F_{\text{ww,CH}_4,\text{default}}$)	Global Warming Potential of Methane (GWP_{CH_4})	Leakage emissions associated with storage of digestate ($\text{LE}_{\text{storage},}$) (tCO ₂)
Sep-21	3,075.19	0.6	0.00067	1.24	0.15	28	6
Oct-21	39,649.80	0.6	0.00067	15.94	0.15	28	67
Nov-21	85,341.00	0.6	0.00067	34.31	0.15	28	145
Dec-21	170,351.72	0.6	0.00067	68.48	0.15	28	288
Sum/Avg 2021	298,417.71	0.6	0.00067	119.96	0.15	28	506
Jan-22	177,312.58	0.6	0.00067	71.28	0.15	28	300
Feb-22	177,232.45	0.6	0.00067	71.25	0.15	28	300
Mar-22	218,215.85	0.6	0.00067	87.72	0.15	28	369
Apr-22	311,287.78	0.6	0.00067	125.14	0.15	28	526
May-22	339,210.14	0.6	0.00067	136.36	0.15	28	573
Jun-22	300,604.20	0.6	0.00067	120.84	0.15	28	508
Jul-22	313,616.55	0.6	0.00067	126.07	0.15	28	530
Aug-22	307,812.82	0.6	0.00067	123.74	0.15	28	520
Sep-22	319,613.41	0.6	0.00067	128.48	0.15	28	540
Oct-22	369,184.54	0.6	0.00067	148.41	0.15	28	624
Nov-22	391,497.04	0.6	0.00067	157.38	0.15	28	662
Dec-22	375,560.00	0.6	0.00067	150.98	0.15	28	635
Sum/Avg 2022	3,601,147.36	0.6	0.00067	1447.66	0.15	28	6087
Jan-23	381,269.43	0.6	0.00067	153.27	0.15	28	644
Feb-23	383,143.65	0.6	0.00067	154.02	0.15	28	647
Mar-23	421,106.95	0.6	0.00067	169.28	0.15	28	711
Sum/Avg 2023	1,185,520.03	0.6	0.00067	476.58	0.15	28	2002

<i>Total Sum</i> 29/09/2021- 31/03/2023	5,085,085	0.6	0.00067	2,044	0.15	28	8,595
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7.5 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
29-September-2021 – 31-December 2021	2,807	0	506	2,301
01-January-2022 – 31-December 2022	33,941	0	6,087	27,854
01-January-2023 – 31 March 2023	9,930	933	2,002	6,995
Total	46,678	933	8,595	37,150

Comparison of ex-ante emission reductions and actual emission reduction is given in the table below.

Ex-ante emissions reductions /removals	Achieved emissions reductions/removals	Percent difference	Justification for the difference
53,929 tCO₂e.	37,150 tCO ₂ e	-45.2%	Ex-ante calculations are much higher than ex-post emission reductions. The reason behind is that the project could not be operational and generate electricity for a while although the plant is commissioned. Moreover, the plant is slowly increased its generated electricity as well as methane destruction. It is expected that the percent difference will be lower in following monitoring periods since the Project approximately reached its full capacity.

APPENDIX 1: MONTHLY ELECTRICITY GENERATION DATA

YILMAZLAR BPP EMISSION REDUCTION AMOUNT (27/06/2021 - 31/03/2023) as per AMS-I.D.			
Months	Net electricity supplied to the grid[MWh]	Baseline emissions or baseline net GHG removals by sinks (tCO ₂ e)	Emission reductions or net anthropogenic GHG removals by sinks (tCO ₂ e)
<i>Sep-21</i>	19.91	11	11
<i>Oct-21</i>	64.27	35	35
<i>Nov-21</i>	145.68	80	80
<i>Dec-21</i>	304.57	169	169
Sum 2021	534.43	295	295
<i>Jan-22</i>	310.97	172	172
<i>Feb-22</i>	315.11	174	174
<i>Mar-22</i>	415.83	230	230
<i>Apr-22</i>	559.78	310	310
<i>May-22</i>	604.61	335	335
<i>Jun-22</i>	547.15	303	303
<i>Jul-22</i>	569.66	316	316
<i>Aug-22</i>	548.74	304	304
<i>Sep-22</i>	606.53	336	336
<i>Oct-22</i>	712.51	395	395
<i>Nov-22</i>	619.55	343	343
<i>Dec-22</i>	648.85	360	360
Sum 2022	6,459.28	3,578	3,578
<i>Jan-23</i>	653.29	362	362
<i>Feb-23</i>	644.33	357	357
<i>Mar-23</i>	700.33	388	388
Sum 2023	1,997.95	1,107	1,107
Total Sum 29/09/2021-31/03/2023	8,992	4,980	4,980

APPENDIX 2: NET ESTIMATED EMISSION REDUCTIONS PER METHODOLOGY

Year		Baseline Emissions from methane (AMS-III.D.)	Baseline Emissions from electricity use (AMS-I.D.)	Total Baseline Emissions	Project Emissions	Leakage	Net Reduction
29/09/2021	31/12/2021	9,944	1,654	11,598	662	1,802	9,134
01/01/2022	31/12/2022	39,028	6,495	45,523	2,597	7,071	35,855
01/01/2023	31/12/2023	39,028	6,495	45,523	2,597	7,071	35,855
01/01/2024	31/12/2024	39,028	6,495	45,523	2,597	7,071	35,855
01/01/2025	31/12/2025	39,028	6,495	45,523	2,597	7,071	35,855
01/01/2026	31/12/2026	39,028	6,495	45,523	2,597	7,071	35,855
01/01/2027	31/12/2027	39,028	6,495	45,523	2,597	7,071	35,855
01/01/2028	28/09/2028	29,083	4,840	33,923	1,936	5,269	26,718
TOTAL		273,195	45,464	318,659	18,180	49,497	250,982
Annual Average		39,028	6,495	45,523	2,597	7,071	35,855